Academic institution

Industrial host

KINGDOM OF MOROCCO

MOULAY ISMAIL UNIVERSITY

NATIONAL SCHOOL OF ARTS AND CRAFTS – MEKNES

INTRODUCTORY ENGINEERING INTERNSHIP REPORT

Industrial Engineering Department

Industrial Engineering – Operational Excellence

First Year of the Engineering Cycle

Digitalization and Design of an Intelligent Supervision Solution for the Soluble MAP Production Process

INDUSTRIAL AI · PROCESS ENGINEERING · DECISION SUPPORT

Internship at OCP Group – Jorf Lasfar

Internship Period: 9 July–9 August 2026

Prepared by:

Chaal Abdelkarim

Industrial Engineering student

Industrial Supervisor:

Mr Mouad CHERIFI

OCP Engineer, Soluble MAP Production Manager

Academic Year: 2025–2026

Acknowledgements

I would first like to thank **Mr Mouad CHERIFI** for supervising this internship and for helping me connect what I observed in the plant with a clear engineering problem. His explanations of the soluble MAP process gave the project its practical direction.

I also thank the **OCP Jorf Lasfar teams** for their welcome and for the time they gave me to understand the production environment and its supervision practices. I am grateful to **ENSAM Meknes and its teaching staff** for the methods and technical foundation used throughout the work.

Finally, I thank **Doha Hailassi** for helping me navigate the company environment and understand its industrial context during the first days of the internship.

Abstract

Final soluble monoammonium phosphate (MAP) moisture is confirmed by laboratory analysis approximately every two hours, while dryer conditions evolve continuously. This internship investigates how that information delay can be reduced without replacing the laboratory or interfering with process control.

I developed an advisory supervision prototype that connects a moisture soft sensor, a One-Class Support Vector Machine (One-Class SVM) novelty detector, transparent diagnostic rules, PostgreSQL and a two-page Power BI dashboard. The data path preserves the difference between fast process observations and sparse laboratory measurements. Process snapshots are shifted by residence time, only strictly previous laboratory context enters a prediction, Python executes all preprocessing and models, and Power BI is limited to visualization through DirectQuery.

The supplied deterministic source contains 92 days of synthetic data: 1,589,760 process rows and 1,104 sparse laboratory observations. On the untouched 165-sample chronological TEST tail, the selected Ridge model obtains MAE 0.001069, RMSE 0.001403 and $R^2 = 0.824530$. It reduces MAE by 26.8% and RMSE by 24.1% relative to carrying forward the previous laboratory moisture. The One-Class SVM has a 5.925% validation novelty flag rate and responds strongly to the held-out steam and airflow disturbances, although it does not flag the feed-surge episode.

These results support the feasibility of the prototype, not plant-wide performance. The five-second rhythm is a replay and visualization choice, the dashboard demonstration uses disclosed injected scenarios, and no autonomous action is implemented. Laboratory moisture remains the quality reference. The proposed next step is a read-only shadow validation using representative OCP historian and laboratory data.

Keywords: Industrial Engineering; soluble MAP; rotary dryer; soft sensor; novelty detection; PostgreSQL; Power BI; decision support.

Contents

List of Figures

List of Tables

List of Abbreviations

AI	Artificial Intelligence
BI	Business Intelligence
CSV	Comma-Separated Values
DOCX	Microsoft Word Open XML Document
ENSAM	National School of Arts and Crafts (*École Nationale Supérieure d'Arts et Métiers*)
IEC	International Electrotechnical Commission
JSON	JavaScript Object Notation
MAE	Mean Absolute Error
MAP	Monoammonium Phosphate
OCP	OCP Group
OCSVM	One-Class Support Vector Machine
PCS 7	Process Control System 7
PDF	Portable Document Format
RBF	Radial Basis Function
RMSE	Root Mean Squared Error
R^2	Coefficient of Determination
SHA-256	Secure Hash Algorithm, 256-bit
SQL	Structured Query Language
SVM	Support Vector Machine
TMDL	Tabular Model Definition Language
UTC	Coordinated Universal Time

General Introduction

During my internship at OCP Group's Jorf Lasfar site, one supervision gap stood out: the soluble monoammonium phosphate (MAP) dryer changes continuously, but final moisture is confirmed by the laboratory only about once every two hours. The laboratory result remains the quality reference, yet it inevitably describes material sampled in the past. The practical question behind this project was therefore simple: *can the information already available around the dryer be organized into a useful view between two laboratory results?*

I approached that question as an Industrial Engineering problem rather than as a model-building exercise. The Artificial Intelligence (AI) layer had to connect process understanding, time alignment, model validation, data storage and an interface that an operator could read quickly. It also had to preserve the boundary between evidence and authority: a prediction is not a new measurement, and a ranked diagnostic clue is not proof of root cause.

The implemented prototype has four connected functions. A Ridge soft sensor estimates final moisture from residence-adjusted process conditions and the most recent strictly previous laboratory context. A One-Class SVM compares each process observation with a synthetic reference regime. A transparent diagnostic layer ranks unusual variables and proposes checks. PostgreSQL then stores the traceable outputs, while Power BI visualizes them through an operations page and an investigation page. Python runs every model; Power BI does not.

The final source is a supplied deterministic 92-day synthetic dataset because sufficient representative plant history was not available. Its process rows lie on a five-second grid chosen for reproducible replay and visualization, while laboratory fields remain sparse at approximately two-hour intervals. This rhythm is not presented as the confirmed native sampling frequency of the plant historian. The software is not connected to the Siemens Process Control System 7 (PCS 7), does not write to a controller and remains advisory.

Three engineering questions guide the report:

1. Can a causally aligned model improve on simply carrying forward the last laboratory moisture value?
2. How does the novelty detector respond to labelled synthetic disturbances outside its training interval, and what does it miss?
3. Can the complete information chain remain traceable and responsive enough for a credible control-room demonstration?

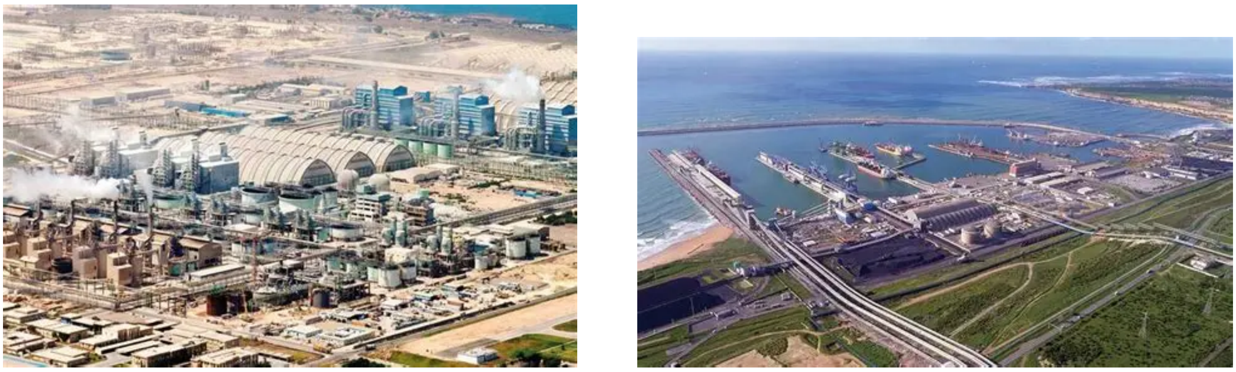
The report follows the same order. Chapters 1 and 2 establish the industrial and physical context. Chapter 3 turns the observation into objectives and measurable prototype criteria. Chapter 4 explains the source data, causal alignment and feature construction. Chapter 5 evaluates the moisture and novelty branches. Chapters 6 and 7 describe the runtime and dashboard, and Chapter 8 brings together the results, limitations and next steps.

CHAPTER 1

Industrial Context and Internship Mission

1.1 OCP Jorf Lasfar and the project setting

OCP Group operates an integrated phosphate value chain, from extraction to transformation and plant-nutrition products. At Jorf Lasfar, production units interact with acid and fertilizer facilities, utilities, storage and port logistics [1, 2]. For this internship, the relevant point is not the corporate scale itself. It is the need to coordinate material, energy, equipment condition and quality information across a continuous industrial system.



(a) Integrated processing platform.

(b) Port and logistics interface.

Figure 1.1: Jorf Lasfar industrial context.
Source: Supplied ENSAM/OCP organization report [3]; contextual images, not evidence of prototype deployment.

My fieldwork focused on the soluble MAP installation. I followed the production sequence from reactant preparation and neutralization to crystallization, centrifugal separation, drying, cooling and handling. The PCS 7 screens provided a useful view of how these stages are supervised. They informed the process analysis in this report, but the prototype has no live PCS 7 connection and does not reproduce its control logic.

1.2 Observed supervision problem

Process variables are available continuously in the control environment, whereas final moisture, density and product temperature are confirmed by periodic laboratory analysis. The last laboratory value is authoritative for the sampled material, but it becomes progressively older while the process continues to change. This is the gap addressed by the project.

The useful digital response is not to hide that age. The interface must state whether a number is a current process observation, a laboratory measurement, a model estimate or a diagnostic interpretation. Timestamp, data freshness and model version are therefore part of the functional design, not optional technical metadata.

1.3 Mission, contribution and boundary

The mission was to design and demonstrate an intelligent supervision architecture around dryer moisture and abnormal-operation visibility. I carried out the process-to-data translation, causal time alignment, feature design, model comparison, anomaly interpretation, database integration, runtime checks and dashboard information design. Python performs the analytical work, PostgreSQL provides traceable persistence, and Power BI presents the result.

During the internship, I built the prototype layer around the existing process, PCS 7 supervision context and laboratory practice. My work covered the process-to-data translation, model workflow, traceable storage and operator-facing dashboard. A convincing interface should still not be mistaken for a deployed plant system.

The deliverable remains a controlled proof of concept. It does not write to equipment, automate conformity decisions, replace laboratory control or claim industrial-scale accuracy. Future use would require approved historian access, tag and clock reconciliation, representative campaigns, cybersecurity review, shadow validation and operating procedures jointly owned by OCP stakeholders.

Design requirements taken from the mission

Every displayed value retains a timestamp and provenance; stale or incomplete states remain visible; anomaly evidence leads to a verification prompt; model and feature versions are traceable; and the complete analytical layer remains outside the control loop.

The following chapter develops the process basis behind these requirements, with particular attention to the dryer, its material delay and the measurements available to the prototype.

CHAPTER 2

Soluble MAP Production Process

2.1 From prepared reactants to conditioned crystals

I reconstructed the soluble-MAP sequence from the supplied process report [4], my field observations [5], selected on-site photographs and the project-specific MAP SOLUBLE PCS 7 screens. The photographs establish physical context, while the screens show how the process is supervised. I used both to understand the production sequence and kept the model boundary focused on the dryer variables available in the repository.

The simplified sequence in fig. 2.1 starts with phosphoric-acid pretreatment and ammonia vaporization. Neutralization forms ammonium phosphate solution, buffer reservoirs stabilize the downstream feed, concentration and vacuum crystallization form crystals, centrifugation removes mother liquor, and drying removes residual water before cooling and conditioning. Recycle streams and intermediate storage couple the stages, so a final-moisture deviation cannot automatically be attributed to one dryer component.

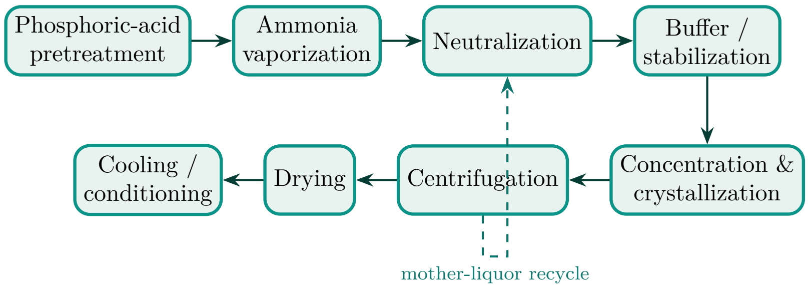


Figure 2.1: Overall soluble MAP production flow, simplified for supervision analysis.
Source: Author's synthesis from [4, 5] and the supplied MAP SOLUBLE PCS 7 screens.

How to read the PCS 7 evidence

The screen title identifies the process stage, while equipment symbols and line connections support a qualitative reading of material flow and supervision. The report uses these images as process context and does not infer unverified tag mappings or operating limits from them.

2.2 Reactant preparation and upstream environmental control

The cited process description presents phosphoric-acid pretreatment through desulfation, dearsenation, flocculation/decantation and aluminization [4]. These operations influence solution composition before neutralization but are not direct inputs to the current dryer model. The project-specific PCS 7 and on-site evidence therefore receive visual priority.

Liquid ammonia is vaporized before neutralization. The supplied PCS 7 views show vaporization and acid scrubbing as separate supervised utilities. Both remain outside the model-input boundary.

2.3 Neutralization and stabilization

The principal reaction may be represented by

$$H_3PO_4 + NH_3 \rightleftharpoons NH_4H_2PO_4, \tag{2.1}$$

with further ammoniation capable of producing diammonium phosphate depending on ratio and conditions [4]. Neutralization is exothermic. Acid/ammonia balance, pH, temperature, agitation and recycled mother liquor therefore belong to one interacting operating context. These concepts are supported by the process description; the current prototype has no direct neutralization columns and does not claim to reconstruct the PCS 7 control loops.

The full screen in fig. 2.2 is identified by its interface title as the neutralization stage. It shows three agitated vessel symbols, their associated drives and pumps, and connections for reagent, condensate, mother liquor and gas services. The detail crop makes the vessel train easier to read. I did not assign the vessels to on-site photographs because the available images do not show reliable identifiers.

Buffer reservoirs absorb short-term variation and decouple neutralization from downstream processing. The screen in fig. 2.3 presents four agitated reservoirs with connected pumps and process lines. Its relevance is temporal: buffering delays and mixes disturbances, so an upstream change need not appear at the final product at the same timestamp.

2.4 Crystallization and centrifugal separation

Crystallization requires supersaturation. In the observed process, circulation, vacuum operation and thermal exchange support concentration and crystal formation. The two supplied PCS 7 views identify the 300 and 400 second-effect crystallization lines. Their side-by-side presentation in fig. 2.4 shows that the operating context is distributed across two lines rather than represented by a single vessel.

The suspension leaving crystallization is sent to centrifugal separation. The corresponding PCS 7 screen shows two centrifuge symbols, feed and service connections, motors and downstream collection. Separation removes much of the mother liquor before thermal drying. Wetter cake raises the evaporation duty imposed on the dryer, so upstream load remains a useful diagnostic hypothesis rather than a confirmed dryer failure.

2.5 Drying-stage process zoom

The dryer receives wet crystals whose residual liquid must be reduced before final handling. Inside a cascading rotary dryer, solids movement, gas–solid contact and simultaneous heat and mass transfer determine the result; the equipment-specific flight geometry and transport behaviour also matter [16]. Hot drying air supplies the transfer potential, while wet loading, gas-flow condition and residence determine the moisture that can be removed. Cooling then reduces product temperature. Figure 2.5 connects this physical reading to the variables represented in the repository.

Figure 2.6 connects three views of the drying stage. Two on-site photographs show the structure and rotary shell; the PCS 7 view shows the supervised dryer train and downstream path. Together they provide physical and operational context for the variables used by the prototype.

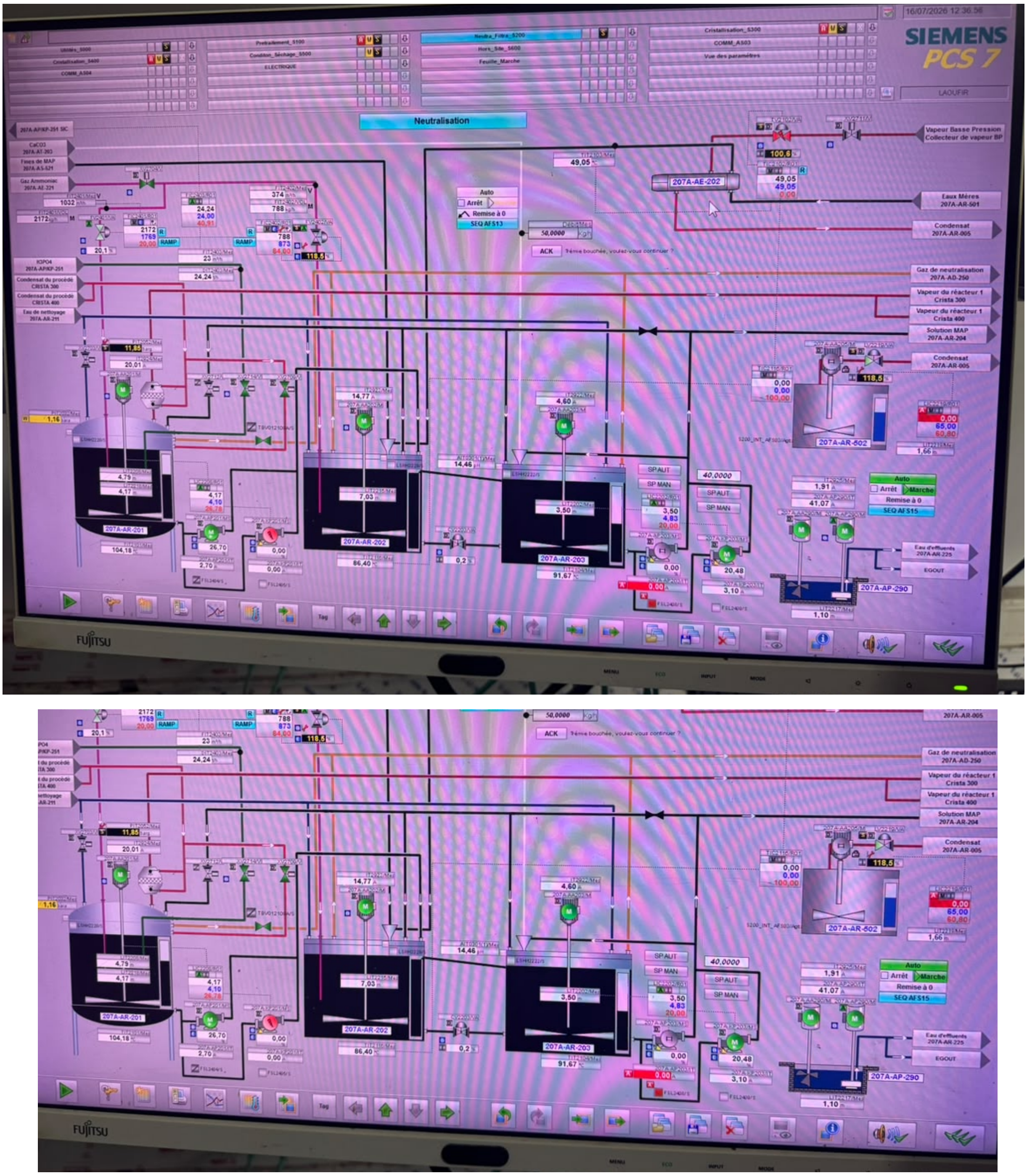


Figure 2.2: PCS 7 neutralization screen (top) and detail crop centred on the three agitated vessels (bottom).
Source: MAP SOLUBLE PCS 7 photograph supplied by the author.

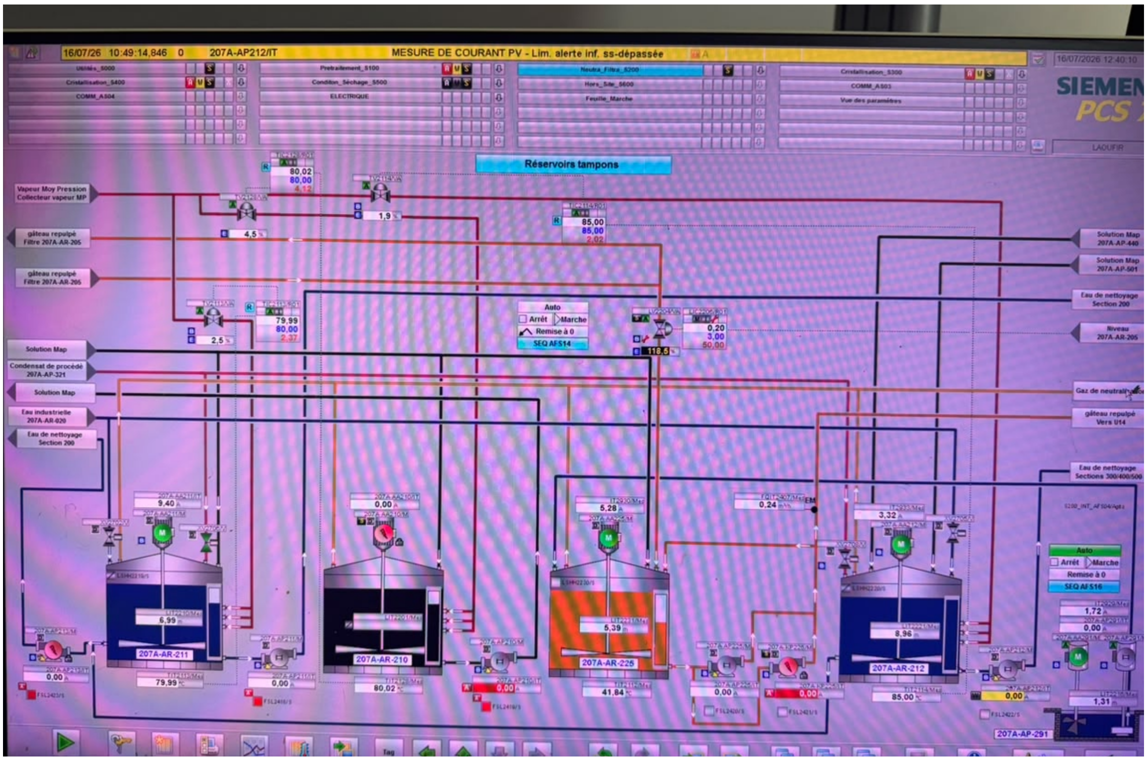
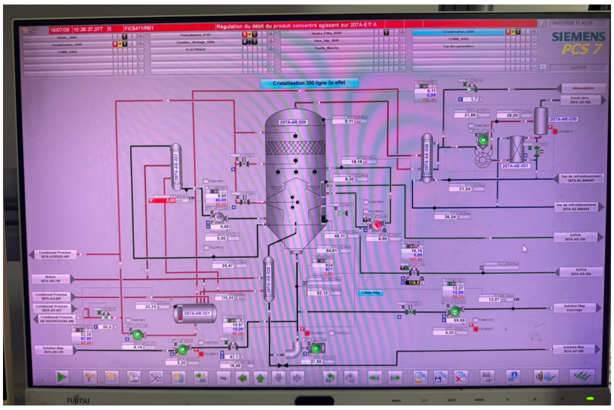
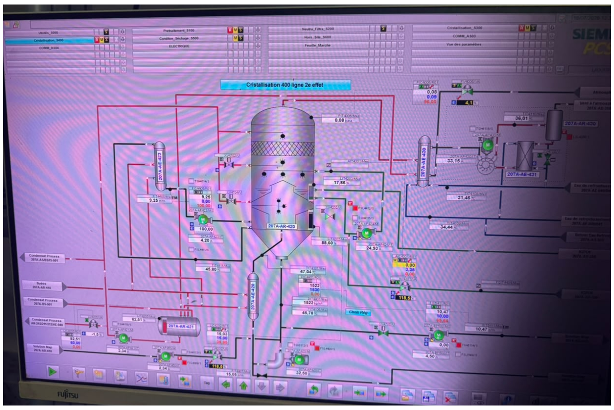


Figure 2.3: PCS 7 buffer-reservoir synoptic used to explain process decoupling and material delay.
Source: MAP SOLUBLE PCS 7 photograph supplied by the author.



(a) Crystallization 300, second effect.



(b) Crystallization 400, second effect.

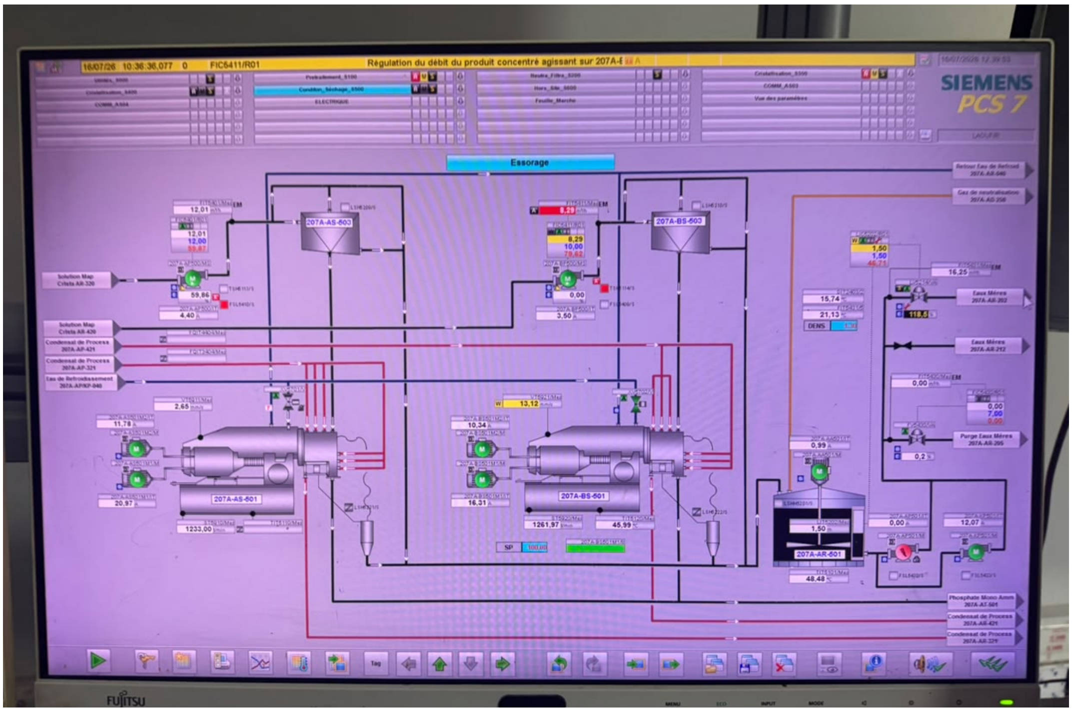


Figure 2.4: PCS 7 views of the two crystallization effects (top) and centrifugal separation (bottom).
Source: MAP SOLUBLE PCS 7 photographs supplied by the author.

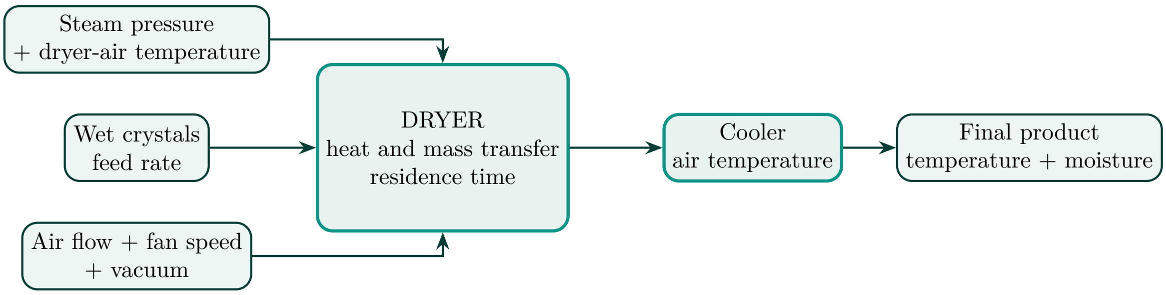


Figure 2.5: Drying-stage engineering zoom and the principal variables represented by the prototype.
Source: Author's synthesis from the repository data dictionary and feature metadata [7].

(a) On-site drying-section structure and process ductwork.

(b) On-site view of the rotary dryer shell.

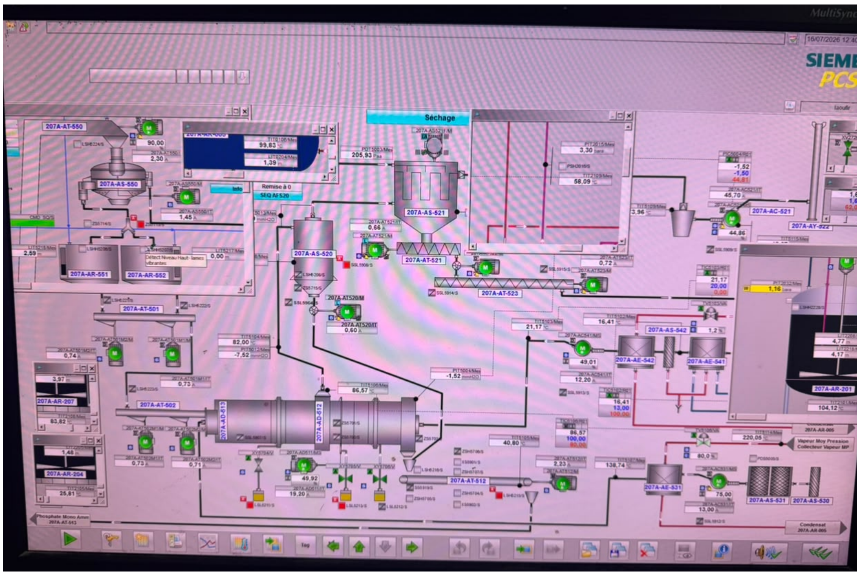


Figure 2.6: Selected on-site drying views (top) paired with the PCS 7 drying supervision screen (bottom). *Source: On-site photographs and MAP SOLUBLE PCS 7 screen supplied by the author.*

The project defines nine fast process inputs: dryer-air temperature, cooler-air temperature, air-flow rate, wet-product feed rate, product-inlet temperature, residence time, vacuum, steam pressure and fan speed. Product density, final product temperature and final moisture are sparse two-hour quality variables in the demonstration contract [7]. Engineered balances such as air per feed, temperature differences and thermal exposure encode interactions without claiming to be first-principles energy meters.

Downstream evidence is selected just as conservatively. Figure 2.7 shows one clear transfer installation and one storage view where soluble-MAP packaging is visibly identifiable. Repeated hopper close-ups, lower-quality transfer angles and packaging-film images that could not be tied specifically to soluble MAP were excluded.

(a) On-site hopper and enclosed solids-transfer equipment; no equipment tag is asserted.

(b) Palletized soluble-MAP product in the finished-product storage area.

Figure 2.7: Selected on-site evidence for downstream solids handling and product storage.
Source: Original on-site photographs supplied by the author; aspect ratios preserved and technical content unaltered.

2.5.1 Mass- and heat-transfer interpretation

Moisture must first be expressed on a consistent basis. If x_{wb} is the wet-basis mass fraction and X_{db} is the dry-basis water-to-dry-solid ratio, then

$$X_{db} = \frac{x_{wb}}{1 - x_{wb}}, \qquad x_{wb} = \frac{X_{db}}{1 + X_{db}}. \tag{2.2}$$

If $\dot{m}_s$ is dry-solid flow, the evaporation demand is then

$$\dot{m}_{evap} = \dot{m}_s \left(X_{in} - X_{out} \right). \tag{2.3}$$

On the gas side, an approximate available duty could be written as

$$\dot{Q}_{available} = \dot{m}_g \left(h_{g,in} - h_{g,out} \right), \tag{2.4}$$

and related to evaporation and sensible heating through

$$\dot{Q}_{available} \approx \dot{m}_{evap} \lambda_v + \dot{m}_s c_{p,s} \left(T_{s,out} - T_{s,in} \right) + \dot{Q}_{losses}. \tag{2.5}$$

This framework explains the direction of the main influences, but the current dataset does not contain inlet wet-cake moisture, dry-solids mass flow, gas humidity/enthalpy or heat losses. A defensible heat balance therefore cannot be closed. The variables called thermal load, steam–temperature interaction and thermal exposure later in the report are statistical proxies, not measured heat duty or energy consumption. Detailed rotary-dryer modelling would also require dryer-specific transport and transfer parameters followed by experimental validation [16].

The incomplete balance still clarifies the supervision problem. More feed or wetter centrifuge cake increases evaporation demand; inadequate thermal supply or gas transfer can raise final moisture; and a change in residence affects exposure. The five-second grid used later is a deliberate replay choice, not a claim about the native frequency of plant measurements.

Steam pressure and dryer-air temperature act as prototype indicators of thermal supply; air flow, fan speed and vacuum describe the air/exhaust path; wet-product feed and residence time describe loading and exposure. Changing one variable is not an automatic recommendation. Higher heat or air may affect energy use, final temperature or other constraints, while feed reduction affects production. The model therefore produces evidence and verification prompts only; PCS 7 and approved OCP procedures remain authoritative.

2.6 From the physical process to the analytical layer

The analytical boundary is concentrated on the dryer. Neutralization, buffering, crystallization and centrifugation explain upstream chemistry, delay and wet-cake loading, but no direct tag mapping from those PCS 7 screens is claimed. The implemented dataset uses verified dryer concepts: thermal supply, drying air/exhaust, product loading and exposure, cooling, and sparse final-quality observations.

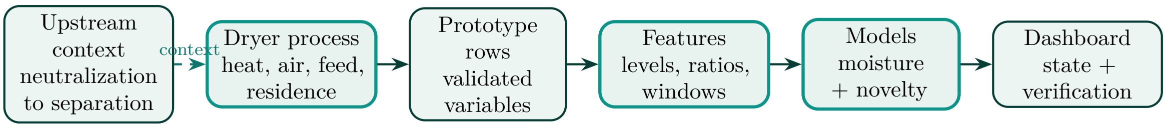


Figure 2.8: Process-to-data-to-AI chain used in the project.
Source: Author's synthesis from the repository data contract and implemented architecture [7, 6].

Thermal variables and air/feed conditions enter the moisture and anomaly paths; residence time also supports temporal alignment. Final moisture remains the sparse supervised target. Density and final product temperature enter the moisture model only as the most recent strictly previous laboratory results; they are not treated as five-second measurements. The diagram communicates the implemented chain without implying that every upstream stage is a model input or that a PCS 7 screen tag equals a repository column.

2.7 Moisture supervision and temporal interpretation

Final moisture is selected because it is strongly influenced during drying, is directly linked to product quality, and can change before the next two-hour laboratory result. This does not imply a single cause. Thermal supply, drying-air delivery, exhaust condition, feed loading, upstream wetness, effective residence and cooling context can interact.

Process monitoring also involves delay. Neutralization and crystallization disturbances do not appear instantaneously at the final product because reservoirs, circulation, crystallizers, centrifuges and the dryer mix and delay material. The available repository does not track each parcel, so the supervised alignment explicitly accounts for dryer residence time and leaves additional transport delay configurable. This corrects the most direct temporal mismatch without claiming a complete dynamic model of the plant.

The PCS 7 and Power BI roles are complementary. PCS 7 is suited to equipment state and control-loop detail. The analytical layer combines longer time horizons, laboratory age, model validation and cross-variable evidence. When an event appears in Power BI, the operator returns to PCS 7, laboratory evidence and approved field procedures for verification. The prototype neither writes to PCS 7 nor changes a setpoint.

The laboratory closes the quality loop. A measured result is associated with sampling and analysis, while a prediction uses the residence-adjusted process snapshot and the strictly previous laboratory density and final-product-temperature results. The database preserves both values and evaluates error only at a true sample timestamp. If they disagree, the appropriate investigation covers sampling, time alignment, input quality, model support and process state; the prediction never overwrites the laboratory record.

On-site and supervision evidence boundary

The selected photographs document physical equipment, product handling and the observed PCS 7 supervision context. They do not prove live prototype deployment or a one-to-one mapping between photographed equipment, screen tags and repository fields.

CHAPTER 3

Problem Statement, Objectives and Evaluation Logic

3.1 Turning a delayed measurement into an engineering problem

The difficulty is temporal. An operator can see that air flow, feed, temperature, vacuum or steam pressure has changed, but the next measured moisture value may arrive almost two hours later. The previous laboratory result remains valid for its sample; it is not evidence that the present product is unchanged.

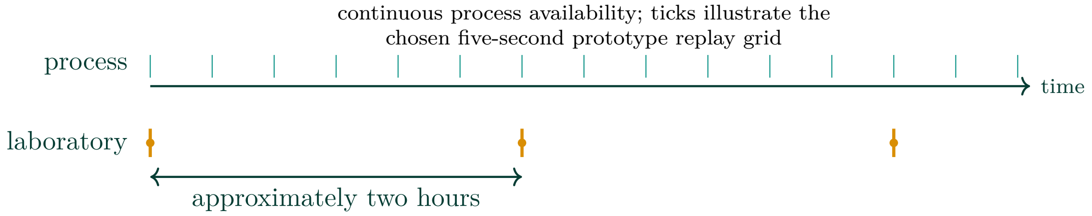


Figure 3.1: Multi-rate timing problem: fast process visibility and sparse laboratory confirmation.
Source: Author's work.

The project inserts an information layer between those two rhythms. It estimates current moisture, identifies departures from a learned reference and organizes the evidence for verification. It does not alter the process control layer. This distinction shaped both the software architecture and the wording of the dashboard.

3.2 Objectives translated into testable criteria

The overall objective is to demonstrate a traceable, near-real-time advisory workflow for soluble MAP dryer supervision. I broke it into four questions and linked each question to evidence that can be checked in the repository.

No approved plant specification, false-alarm cost or response-time baseline was available. I therefore do not invent a production acceptance threshold or savings figure. These audit criteria answer whether the prototype is coherent and informative enough to justify a later shadow study.

3.3 Implemented architecture and responsibility boundary

Python owns preprocessing and every model calculation. PostgreSQL owns persistence and stable Structured Query Language (SQL) views. Power BI filters and visualizes those views; it neither trains nor runs the models. Two replay sources serve different purposes: the canonical held-out TEST interval supports evaluation, while a disclosed two-day dashboard fork provides a visible sequence of demo events. Results from the second source are never substituted for TEST metrics.

My contribution was to connect these parts into one reproducible workflow: causal alignment, model comparison, novelty interpretation, database integration and dashboard information design.

Table 3.1: Prototype objectives and final audit criteria

Objective	Audit criterion	Evidence used
Preserve multi-rate meaning	No future process or laboratory value enters a supervised row; laboratory blanks remain sparse.	Timestamp audit, residence shift, strictly previous lab lookup, 1,103 aligned rows.
Improve moisture visibility	Compare the selected model with a credible persistence baseline on one untouched chronological TEST segment.	Holdout errors, bias, residuals and baseline improvement.
Test process novelty	Evaluate the frozen detector on labelled synthetic episodes outside TRAIN and report its response without treating it as a fault classifier.	Validation novelty flag rate, held-out episode response and ranked contributors.
Demonstrate the chain	Load the frozen artifacts, produce traceable records and expose stable SQL fields to the two Power BI pages.	Full accelerated analytical replay, bounded database smoke test, model/schema versions and dashboard checks.
Protect operator authority	Keep measurement, prediction, anomaly, diagnosis and recommendation visibly distinct.	Advisory wording, freshness, laboratory age and no control output.

Source: Final report audit criteria; they are not pre-registered plant acceptance limits.

Five-second interval: exact meaning

Five seconds is the selected CSV replay, inference-pacing and Power BI visualization rhythm. It is not a confirmed plant historian frequency and does not change the approximately two-hour laboratory reference cycle.

Explicit non-objectives. The project does not connect to live PCS 7, actuate equipment, certify an alarm or safety function, replace the laboratory, or claim production deployment. Every output is advisory and requires verification.

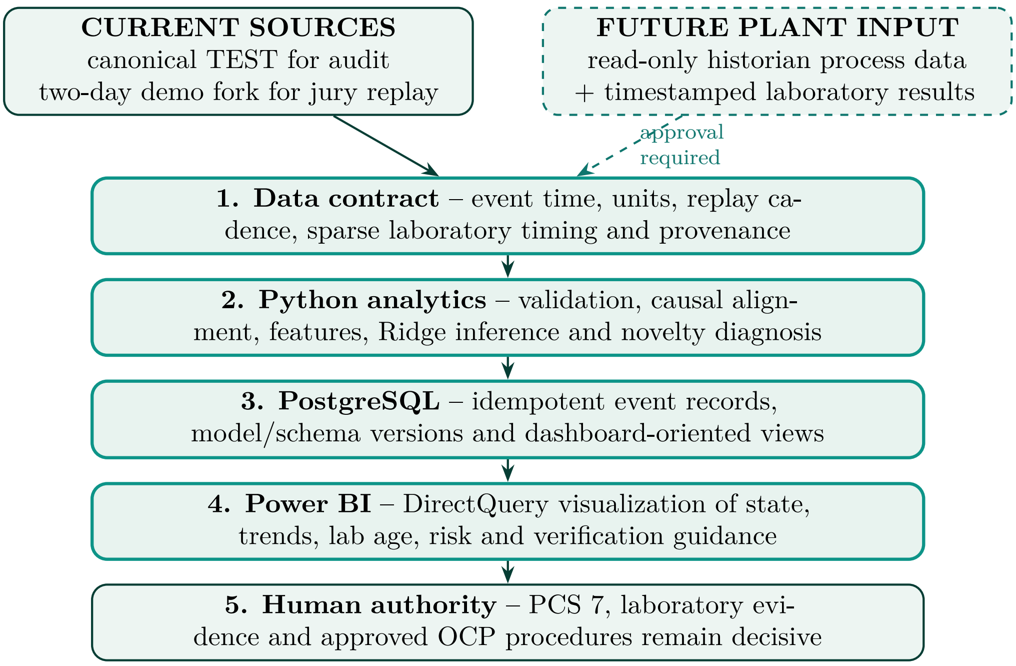


Figure 3.2: End-to-end architecture with implemented and future sources separated.
Source: Author's work.

CHAPTER 4

Data Acquisition, Preparation and Feature Engineering

4.1 Source, provenance and evidence boundary

Sufficient representative plant history was not available for this initiation project, so the final workflow uses one supplied deterministic synthetic Comma-Separated Values (CSV) file. It spans 16 April–16 July 2026 and contains exactly 1,589,760 unique timestamps on a gap-free five-second grid. The three laboratory fields are populated at 1,104 timestamps, equivalent to a regular two-hour pattern. Notebook 01 audits the source; Notebook 02 produces the causal supervised handoff; Notebooks 03 and 04 export the final model artifacts.

The word *deterministic* needs a precise boundary. The file hash matches its manifest, which makes the supplied file reproducible as an artifact. However, the repository does not contain the original generator code, configuration path or random seed. I could therefore audit the data actually present, but I could not reproduce or verify exact generator equations. Relationships reported in this chapter are empirical properties of the CSV, not recovered plant laws or undocumented generation formulas.

Table 4.1: Audited data products and their distinct roles

Data product	Rows	Columns	Lab rows	Cadence	Role
Canonical source	1,589,760	17	1,104	5 s / 2 h lab	Common source for audit and model development.
Aligned supervised handoff	1,103	18	1,103	2 h	Timestamp, 16 inputs and final-moisture target.
Canonical TEST interval	237,600	18	165	5 s / 2 h lab	Frozen-model analytical and database evaluation.
Dashboard demo fork	34,560	18	preserved	5 s / 2 h lab	Two-day jury replay with disclosed injected episodes; not model evaluation.

Source: Direct repository audit generated for this report.

The source labels form 22 synthetic disturbance episodes: five steam dips, five airflow restrictions, four cooling losses, four fan-speed dips and four feed surges. These labels are used for the episode counts and held-out analysis.

Table 4.2: Disturbance episodes found directly in the canonical CSV

Label	Episodes	Interpretation
Steam dip	5	Synthetic reduction in thermal-supply proxy.
Airflow restriction	5	Synthetic drying-air disturbance.
Cooling loss	4	Synthetic downstream cooling disturbance.
Fan-speed dip	4	Synthetic air/exhaust equipment disturbance.
Feed surge	4	Synthetic dryer-loading disturbance.
Total	22	Label audit; not confirmed plant failures.

The prototype uses nine process inputs and three sparse laboratory fields. Missing laboratory timestamps remain blank so that an old result is never presented as a fresh measurement. Vacuum is retained using the convention of the supplied project data; plant tag polarity should be confirmed before industrial deployment [7].

4.2 Exploration without over-interpreting the generator

The process variables occupy compact synthetic ranges with gradual variation and labelled disturbances. Direct calculation gives a fan-speed/air-flow correlation of 0.938, an air-flow/vacuum correlation of 0.979 and a feed/residence correlation of −0.935. At laboratory timestamps, dryer temperature and final moisture correlate at −0.508, while cooler temperature and final product temperature correlate at 0.909. These patterns are useful for checking consistency, but they should not be read as causal plant relationships.

I kept unusual values unless a specific data-quality rule rejected them. In a supervision project, a rare point may be a valid transient, a designed disturbance or an acquisition error. Representative historian data would also need operating mode and sensor-quality information that the synthetic file does not contain.

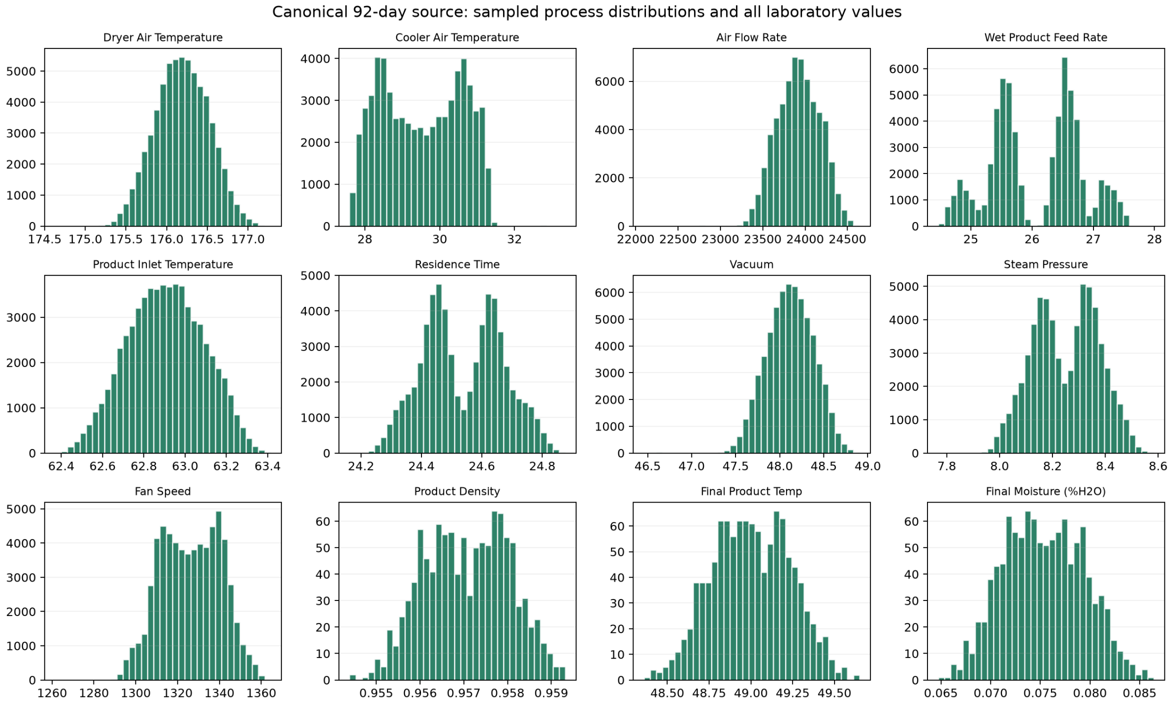


Figure 4.1: Distributions of process and sparse quality variables in the canonical source. *Source: Executed Notebook 01 output.*

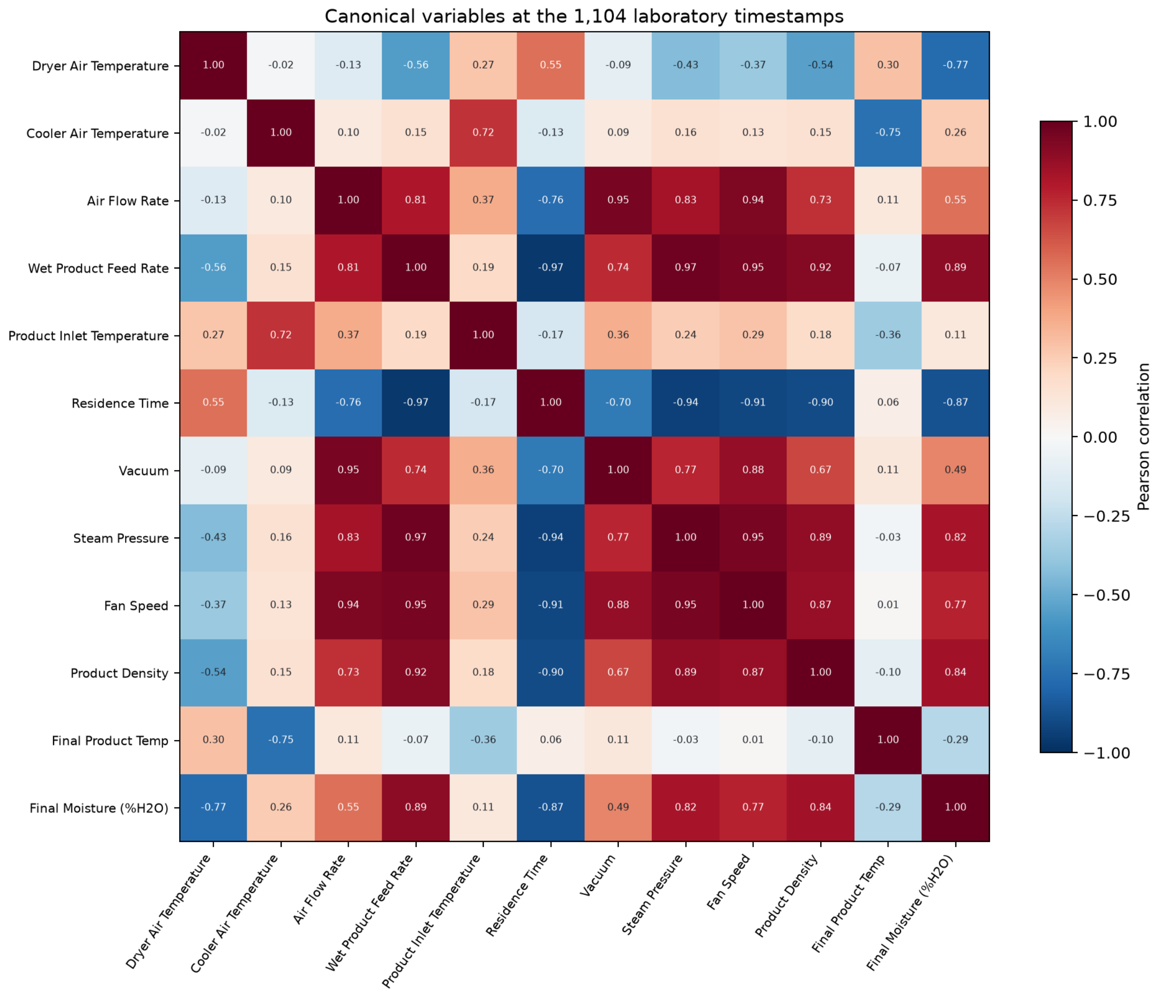


Figure 4.2: Pearson correlation matrix for canonical process and sparse quality variables. *Source: Executed Notebook 02 output.*

4.3 Causal process-to-laboratory alignment

The main leakage risk is not a random missing value; it is time. A laboratory result at t_{lab} describes material that passed through the dryer earlier. The direct process snapshot is selected at

$$t_{effective} = t_{lab} - t_{residence}(t_{lab}) - t_{transport}. \tag{4.1}$$

The generated artifact uses the most recent process row at or before $t_{effective}$ and a configurable transport delay of zero minutes. Density and final product temperature come from the most recent laboratory row strictly before t_{lab}. Final moisture is the target and never an input.

Listing 4.1: Causal snapshot and previous-laboratory selection.

```
effective_time = effective_window_end(
    lab_timestamp, residence, transport_delay_minutes
)
process_row = process.loc[
    process["Timestamp"] <= effective_time
].iloc[-1]
previous_lab = laboratory.loc[
    laboratory["Sample Timestamp"] < lab_timestamp
].iloc[-1]
```

The first laboratory row has no strictly previous laboratory context, leaving 1,103 supervised observations. The chronology is split into 772 TRAIN, 166 VALIDATION and 165 TEST rows. Five expanding folds tune models inside TRAIN, the later VALIDATION block selects the family, and TEST remains closed until selection is complete. Blocked evaluation is appropriate here because random splitting can mix temporally dependent process regimes and make future information indirectly available [14].

4.4 Feature engineering and dimensional meaning

The final moisture contract contains nine direct process values, two prior laboratory values and five interactions. The anomaly contract contains the nine process values plus six instant interactions. Standardization is learned from training data only. The main design choices are summarized below.

Table 4.3: Retained engineered features and what they do not represent

Feature family	Construction	Engineering boundary
Temperature differences	Dryer air minus inlet, cooler air or prior final-product temperature	Thermal contrast; not heat flow.
Air per feed	Air flow divided by wet-product feed	Relative gas/loading index; dimensionless only under the shared volumetric convention.
Steam per feed	Steam pressure divided by wet-product feed	Statistical supply/loading index; units bar/(m^3/h).
Steam–temperature interaction / thermal load	Steam pressure multiplied by dryer-air temperature	Regression interaction; not thermal duty.
Heating index / thermal exposure	Residence time multiplied by dryer-air temperature	Time–temperature exposure proxy; not energy.

Feature names are deliberately qualified because a plausible label can otherwise overstate the underlying physics. The present data do not support a closed mass/energy balance. At runtime, protected denominators and required-input checks produce a `DATA_QUALITY` or unavailable state rather than a silent infinity or improvised prediction. The saved schemas lock feature names, order and versions for both models.

Data conclusion

The canonical source is adequate for reproducible software and method testing. Its size does not remove the central limitation: only 1,103 rows contain supervised moisture targets, the disturbance labels are synthetic, and the original generation procedure is unavailable.

CHAPTER 5

AI Models: Moisture Prediction and Process Novelty

5.1 Two different questions, two different models

A soft sensor estimates a difficult or slow-to-obtain quality variable from easier measurements. This is a common process-industry role, especially when laboratory data arrive slowly, but it also inherits problems such as multi-rate sampling, collinearity, missing data and drift [12]. In this project, the supervised branch answers: *what final moisture is expected from the available context?*

The second branch asks a different question: *does the current process vector resemble the reference regime?* A One-Class Support Vector Machine (One-Class SVM) learns a boundary around reference observations rather than named failure classes [13]. Its output cannot explain a cause by itself. I therefore added a separate diagnostic layer that ranks deviations and suggests what should be checked.

5.2 Soft sensor: governed experiment

Notebook 03 is the source of truth for model selection and the runtime artifact. It receives 1,103 causally aligned rows with 16 ordered inputs: nine direct process values, five engineered interactions and the strictly previous laboratory density and product temperature. The final moisture at the current laboratory timestamp is the target. The chronological split is 772 TRAIN, 166 VALIDATION and 165 untouched TEST rows.

I evaluate the candidates with mean absolute error (MAE), root mean squared error (RMSE) and the coefficient of determination (R^2). Together, these metrics show average error, sensitivity to larger errors and explained variation.

Within TRAIN, five expanding `TimeSeriesSplit` folds tune each candidate. The later VALIDATION block selects the model family; TEST is opened once after that choice. This preserves chronology and avoids the optimistic mixing that random cross-validation can create in dependent series [14].

Table 5.1: Audited contract of the selected moisture model

Item	Audited value
Source	`MAP_Dryer_Lab_Aligned_16.csv`; 1,103 laboratory-aligned rows.
Chronology	TRAIN 772; VALIDATION 166; TEST 165; no shuffle.
Selection	Expanding five-fold tuning inside TRAIN; family selected by later VALIDATION RMSE.
Pipeline	Standardization followed by Ridge Regression, $\alpha = 10$.
Contract	Exactly 16 original ordered inputs; final moisture excluded from features.
Artifact	`models/5s/quality_moisture_pipeline.joblib`.

Source: Executed Notebook 03 and saved evaluation record [8].

Five model families were compared under the same structure. Ridge obtained the lowest VALIDATION RMSE, 0.0013566 percentage points, narrowly ahead of Elastic Net at 0.0013625 and ordinary Linear Regression at 0.0013671. The Ridge/Elastic Net difference is only about 0.43%, so I treat Ridge as the selected version for this dataset and split, not as a universal winner.

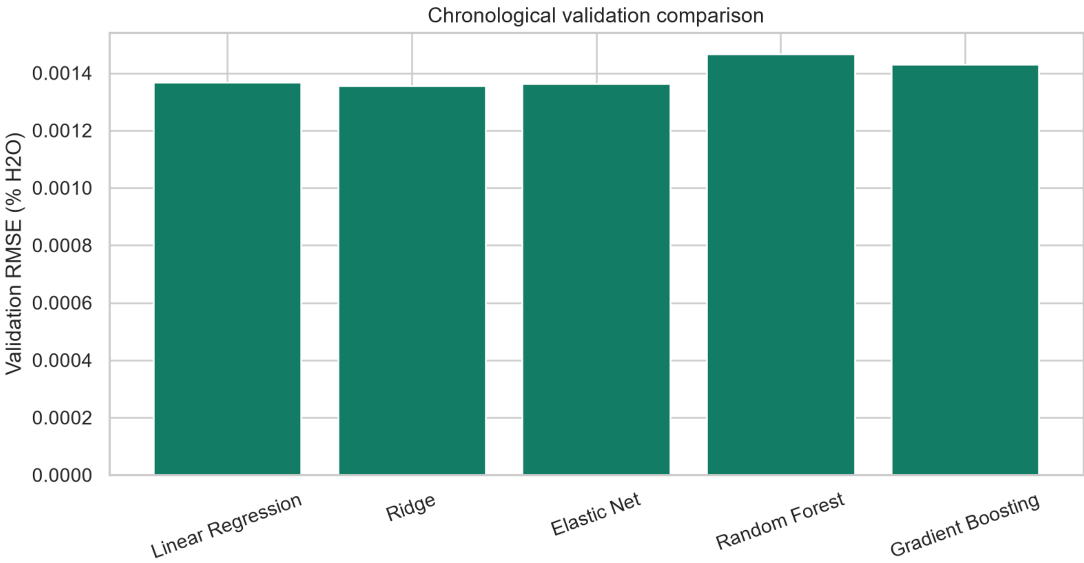


Figure 5.1: Later-segment VALIDATION RMSE used after training-only tuning.
Source: Executed Notebook 03 [8].

5.3 Holdout performance and baseline value

After selection, Ridge is refitted on TRAIN plus VALIDATION and evaluated on the 165-sample TEST tail. It obtains MAE 0.0010686, RMSE 0.0014026, $R^2 = 0.8245295$, bias 0.0000662 and maximum absolute error 0.0056206 percentage points H_2O.

A model score is more informative when compared with a simple operational alternative. I therefore added a persistence baseline that predicts the current moisture using the strictly previous laboratory moisture. On the same TEST rows, persistence gives MAE 0.0014589 and RMSE 0.0018479. Ridge reduces those errors by 26.8% and 24.1%, respectively.

Table 5.2: Untouched TEST comparison with the last-laboratory persistence baseline

Method	n	MAE	RMSE	R^2	Bias
Previous lab moisture	165	0.001459	0.001848	0.695439	0.000012
Selected Ridge	165	**0.001069**	**0.001403**	**0.824530**	0.000066

Source: Independent report audit using the frozen artifact and identical TEST timestamps.

The residual mean is 0.0000662 and the median is 0.0000550; 47.9% of residuals are negative and 52.1% positive. The 5th and 95th percentiles are −0.002170 and 0.002152. A Durbin–Watson statistic of 1.827 and lag-1 residual autocorrelation of 0.084 show limited remaining short-lag structure in this synthetic TEST sequence. Absolute residual and fitted value have correlation 0.054, so no strong level-dependent error pattern is visible. These diagnostics are useful, but 165 sparse targets from one synthetic chronology remain too little evidence for plant acceptance.

Coefficient magnitude is also interpreted cautiously. Air per feed and wet-product feed have the largest absolute standardized coefficients, followed by cooler- and dryer-air temperature. The inputs are correlated, so weight can be redistributed among them; a large Ridge coefficient is not a causal effect.

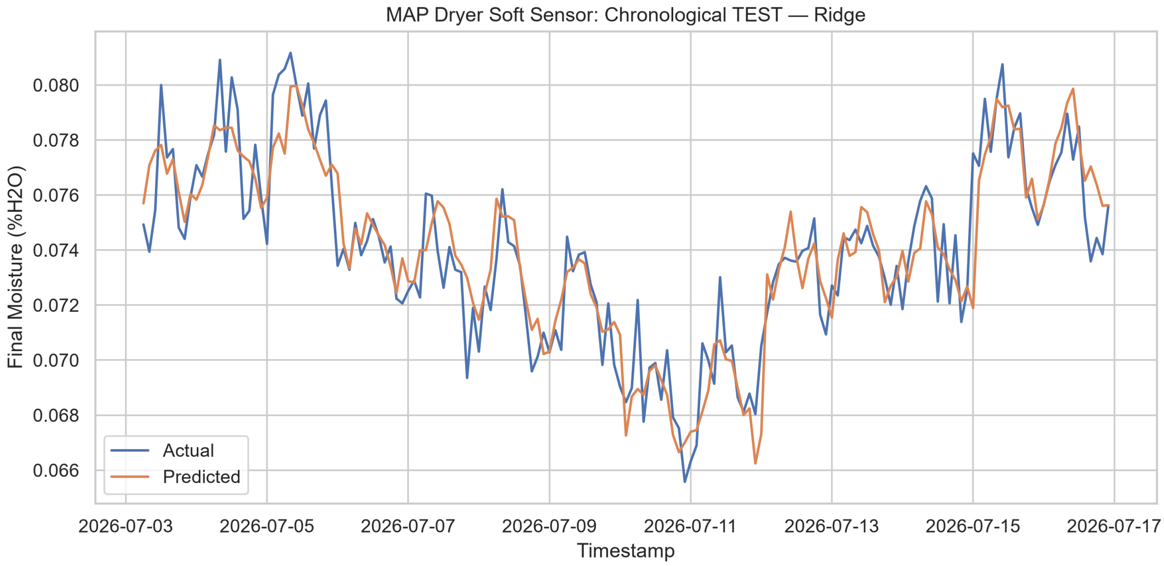


Figure 5.2: Measured and predicted moisture on the untouched chronological TEST interval.
Source: Executed Notebook 03 [8].

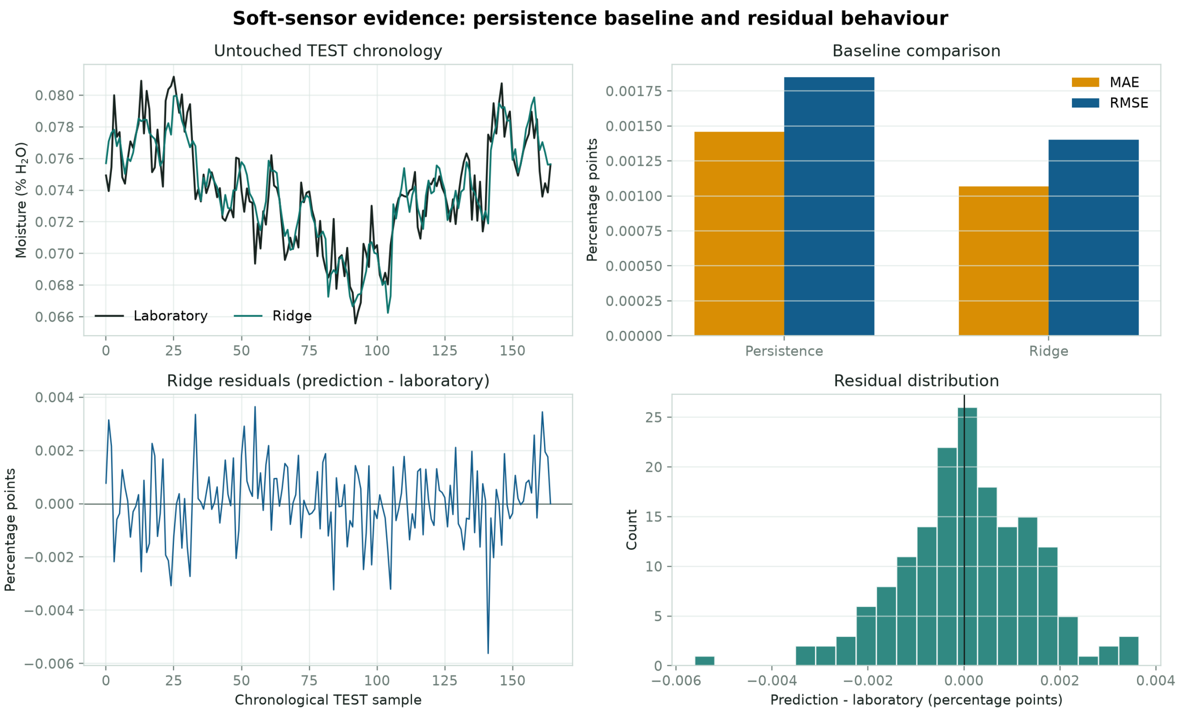


Figure 5.3: Persistence comparison and residual audit for the selected Ridge soft sensor.
Source: Independent report evidence generated from the frozen artifact.

5.4 One-Class SVM for earlier process awareness

Dryer variables evolve together, and an unusual combination can appear before the next laboratory result. I use the One-Class SVM as a process-novelty detector: it compares the current multivariate state with the learned reference region. A novel state prompts investigation; it is not a fault probability, a root-cause classification or a control command.

Notebook 04 exports a radial basis function (RBF) One-Class SVM with `gamma='scale'` and $\nu = 0.02$ [9]. It fits 6,000 sampled rows from the canonical TRAIN period and checks a separate 20,000-row validation sample. The fraction of validation states identified as novel is 5.925%, reported here as the *validation novelty flag rate*.

The raw decision function is a signed model score, not a probability. The service converts it to a zero-to-one display risk for easier reading, with 0.5 at the detector boundary and 0.8 as the visual high-risk level. These values support prioritization on the dashboard; they do not represent failure probability or confidence.

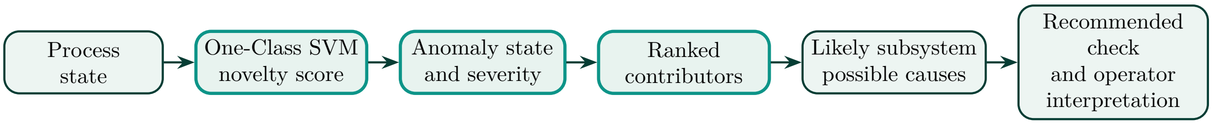


Figure 5.4: Implemented path from process novelty to operator verification.
Source: Author's synthesis of the runtime and diagnostic workflow.

5.5 Held-out process-novelty response

The chronological TEST interval contains 237,600 rows, including 1,620 rows from three labelled synthetic disturbance episodes. The frozen detector flags 6,305 rows in total, or 2.654% of the interval. It flags 60.25% of rows inside the labelled episodes and 2.258% of rows outside them. These are scenario-response and normal-period flag rates, not accuracy or plant fault probabilities.

Table 5.3: Frozen novelty-detector response to the three labelled TEST episodes

Episode	Rows	Flagged	Row response	First flag / interpretation
Feed surge	408	0	0.0%	No flag; peak display risk 0.360.
Steam dip	540	443	82.0%	First raw flag after 290 s; peak risk 0.978.
Airflow restriction	672	533	79.3%	First raw flag after 360 s; peak risk 0.950.

Source: Independent report audit on the held-out canonical TEST interval.

The detector reacts strongly to the steam-dip and airflow-restriction departures, although sensitivity varies with disturbance type and magnitude. It does not flag the feed-surge episode. This is useful design feedback: representative operating modes and better loading-related features matter more than presenting the model as a complete fault classifier.

5.6 From novelty to investigation guidance

The useful contribution is the full diagnostic chain, not only the binary flag. The runtime compares a novel row with the TRAIN-period reference profile, ranks direct-variable deviations, gathers likely-subsystem evidence and presents possible causes with recommended verification. Detection, evidence localization and causal diagnosis remain separate tasks [15].

In a representative dashboard demonstration event, vacuum, air flow and fan speed are the strongest contributors. Their joint movement points to the drying-air and exhaust subsystem, so the interface suggests checking fan demand, damper position, leakage and the vacuum measurement. This is a probable diagnosis for investigation, not a confirmed root cause.

The detector was developed and evaluated on deterministic synthetic process data and should therefore be interpreted as a proof of concept for process-novelty monitoring. Its operating threshold and sensitivity would need to be recalibrated using representative plant history, operating modes and confirmed events before industrial use.

How the anomaly branch is used

The detector acts as an early-warning layer. It highlights a process state, localizes the strongest evidence and leaves verification and action to the operator using PCS 7, laboratory information and normal operating procedures.

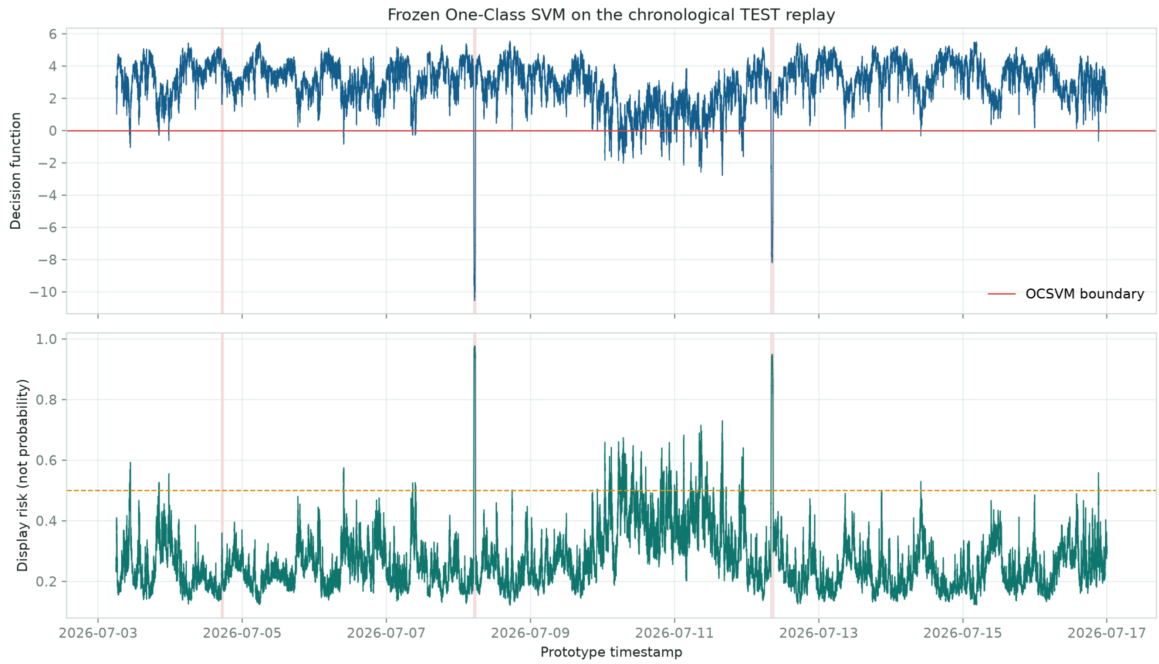


Figure 5.5: One-Class Support Vector Machine (OCSVM) decision function and display risk on the chronological TEST replay.
Source: Frozen One-Class SVM evaluated without refitting.

CHAPTER 6

Real-Time Prototype Architecture

6.1 From notebooks to one governed runtime

The repository separates experimentation from execution. Notebooks 01–04 audit the canonical source, build the causal handoff, select the moisture model and export the anomaly artifacts. Reusable modules under `src` implement alignment, instant features and diagnosis. The `realtime_pipeline` loads the saved artifacts and writes PostgreSQL records. The Power BI project remains a visualization client of that backend [6].

This structure avoids a common prototype failure: showing one model in a notebook and running another in the dashboard. Startup checks compare the ordered feature schemas and model versions. The service loads the same Ridge pipeline and One-Class SVM artifacts evaluated in Chapter 5.

6.2 Two replay modes with separate purposes

The project now uses two explicit sources. The canonical TEST interval supports model and analytical evaluation. It contains 237,600 five-second rows and 165 sparse laboratory observations from 3–16 July 2026. The default jury presentation uses `resources/dashboard_demo/MAP_Dryer_Dashboard_Demo_5s.csv`, a two-day, 34,560-row fork containing eight disclosed injected episodes. Its normal rows and sparse laboratory values are preserved from the source; the injections make short-session dashboard state changes visible.

The distinction is strict. Canonical TEST metrics assess frozen-model behaviour. Dashboard-demo response assesses whether a scripted presentation sequence reaches the intended views. Injected demo episodes are never counted as independent validation evidence.

The five-second cadence belongs to event timestamps and, when enabled, to presentation pacing. An accelerated audit can process those same timestamps without sleeping. Removing the wait changes wall-clock duration, not process chronology or feature alignment.

6.3 PostgreSQL as traceability boundary

The database stores process records, model outputs and ranked contributors under the same event time. Upsert functions use the date/time key so that a retried delivery updates the logical event rather than creating duplicates. One transaction covers the related writes. If persistence fails after inference, the partial cycle can be rolled back.

The overview uses a restricted recent-trend view because an as-of laboratory lookup across every historical point is unnecessarily expensive for a page that refreshes frequently. This is a workload-driven design choice. Stable SQL column names also insulate the Power BI semantic model from implementation details in the base tables.

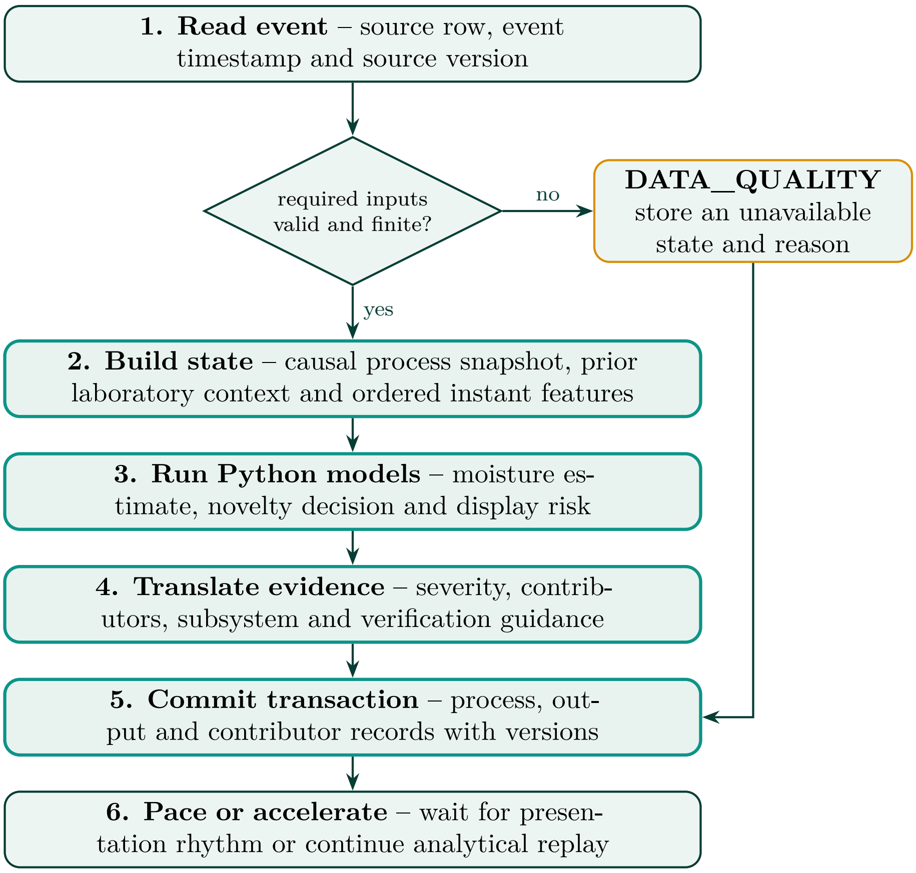


Figure 6.1: Event workflow used by both canonical audit and dashboard demonstration modes.
Source: Author's work.

Table 6.1: Principal PostgreSQL objects and their dashboard roles

Object	Role
`dryer_map`	Nine process fields and three nullable laboratory fields at event grain.
`dryer_model_outputs`	Prediction, risk, state, diagnosis, versions and inference latency.
`dryer_abnormal_variables`	Ranked direct contributor evidence.
`vw_dryer_dashboard_powerbi`	Joined current-grain facts with latest-lab context and validated error.
`vw_dryer_overview_trends_powerbi`	Lightweight recent trend for frequent overview queries.
`vw_dryer_lab_samples`	True laboratory rows and prediction-at-sample error.
`vw_dryer_anomaly_events`	Consecutive anomalous rows grouped into candidate events.
`vw_dryer_contributors_powerbi`	Contributor evidence exposed to the diagnostic page.

Source: Database bootstrap and dashboard-view scripts [10].

6.4 What the timing evidence actually proves

Two performance checks answer different questions. First, an independent accelerated analytical audit ran both frozen models and feature construction over all 237,600 canonical TEST rows. It produced 237,600 moisture predictions and 6,305 anomaly flags in 2.037 s, equivalent to about 116,626 rows/s. Normalized chunk latency averaged 0.00858 ms per row and its 95th percentile was 0.00873 ms per row.

This batch result measures numerical processing only. It excludes database writes, per-row diagnostic text generation, Power BI queries, transport, concurrent users and deliberate pacing. It establishes that the model calculations are not the likely bottleneck; it does not qualify an end-to-end plant service.

Second, the recorded PostgreSQL smoke test processed 1,500 canonical TEST rows and two sparse laboratory observations. Average and maximum cycle latency were 9 ms and 47 ms. That bounded run confirms artifact loading, prior-lab state, transactional writes and SQL-view compatibility. It is not evidence that all 237,600 rows were replayed through PostgreSQL under production load [11].

6.5 Power BI contract and refresh behaviour

The semantic model uses six DirectQuery tables, including a lightweight latest-state view. Power BI sends queries to PostgreSQL at refresh; it does not execute the Ridge or One-Class SVM. Microsoft documents automatic page refresh for DirectQuery and notes that effective cadence depends on query duration, source load, gateway and capacity settings [18]. The configured interval is therefore a presentation target, not a guaranteed service-level agreement.

The latest launcher opens the jury dashboard in a presentation-oriented full-screen workflow and keeps the replay service separate from Power BI rendering. If the user selects a shorter two-second visual interval, source queries can overlap or queue and make the report slower rather than faster. Smoothness depends on completing each query cycle before the next request, so the five-second setting is the stable default.

6.6 Failure visibility and deployment concerns

Missing or non-finite inputs produce `DATA_QUALITY`. Missing shifted history or prior laboratory context produces an unavailable moisture reason. Logs retain event time, source position, model/schema version, state and latency without storing credentials. This makes "the process appears unusual" visibly different from "the analytical chain cannot evaluate the row."

The prototype keeps plant-local timestamps for compatibility. Industrialization would need Coordinated Universal Time (UTC) handling, approved read-only interfaces, managed secrets, network segmentation, recovery procedures and model rollback. Qualification should also measure source age and end-to-end latency under concurrent dashboard load, then test delayed, reordered and missing events.

Architecture conclusion

The complete canonical TEST interval is computationally tractable in accelerated analytical mode, and the bounded database path is coherent. End-to-end five-second operation in an OCP environment remains a future load, resilience and governance test.

CHAPTER 7

Power BI Supervision Dashboard

7.1 A control-room view, not a model host

The dashboard is where the analytical work becomes usable. I designed it around two operator questions: *what is happening now?* and *what should I inspect?* The first page keeps moisture, laboratory age, process state and the main variables together. The second page opens the evidence behind an abnormal event. This separation avoids turning routine monitoring into a dense diagnostic screen.

Power BI does not run either model. It uses DirectQuery against PostgreSQL views populated by the Python service. The semantic model formats values, applies filters and computes display measures; prediction, novelty scoring and diagnostic transformation have already occurred before the row reaches Power BI.

The Power BI project requests five-second automatic page refresh. Actual refresh can be slower because Power BI waits on DirectQuery, local rendering and source availability [18]. The full-screen capture workflow used for the figures below displays a more conservative 60-second presentation label, while the underlying event stream remains five seconds. Neither interval is a confirmed plant historian frequency.

7.2 Page 1: Operations Overview

Figure 7.1 uses the final high-precision layout. The upper cards distinguish predicted moisture, laboratory moisture, validated error, anomaly score, operating status and severity. The main trend preserves the time relationship among prediction, sparse laboratory markers and anomaly points. Critical variables remain visible below, while the right-hand assessment and guidance panels translate the latest event into an operator-readable summary.

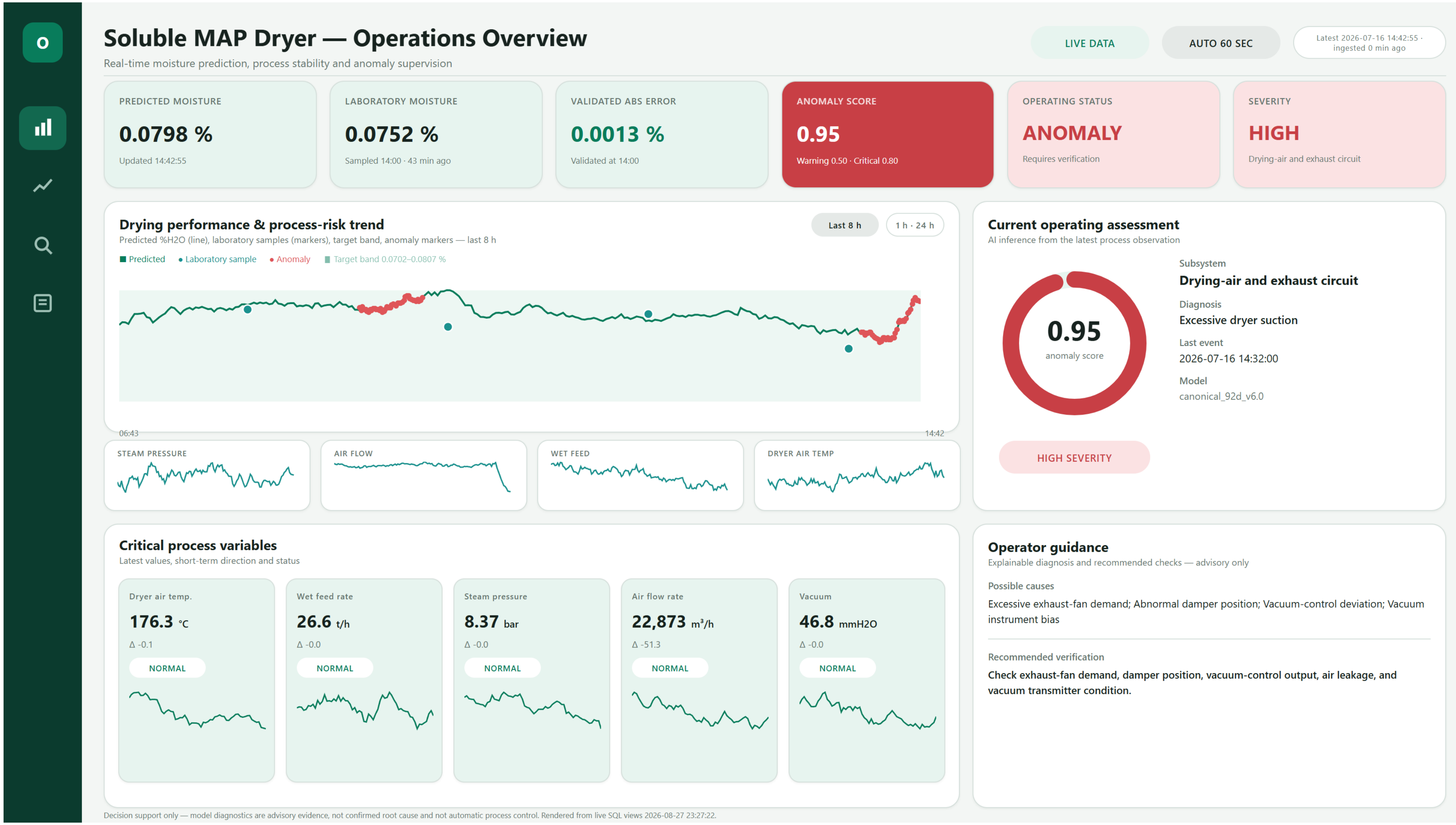


Figure 7.1: Final Operations Overview capture with high-precision moisture values, process trends and advisory context.
Source: Current SQL-backed full-screen dashboard capture; Author's work.

The example shown is an injected dashboard-demo event, not a canonical TEST result or a live plant state. In the capture, 0.0798% predicted moisture is shown separately from the last 0.0752% laboratory result, and the validated absolute error card refers to a true sample timestamp. The "LIVE DATA" pill means that the capture queried the current PostgreSQL view; the source behind that view is still the controlled replay. This distinction is stated here because the interface itself is designed for a future live source.

Three display rules protect interpretation:

- predicted moisture is a continuous model estimate;
- laboratory moisture is a sparse reference accompanied by sample time or age; and
- validated error is calculated only where a represented laboratory sample and prediction coexist.

Status colour supports scanning but never carries meaning alone. Green, amber, red and grey are paired with text labels. Numeric precision is deliberately greater for moisture and error than for larger process measurements; this prevents a useful small change from disappearing through formatting while avoiding false precision on every variable.

7.3 Page 2: Diagnostics and Investigation

The diagnostic page in fig. 7.2 begins with a novelty timeline and a selected event. Ranked contributors show which direct variables depart most from their reference profile. The likely subsystem, possible causes and recommended checks appear beside the signature rather than behind a technical model score.

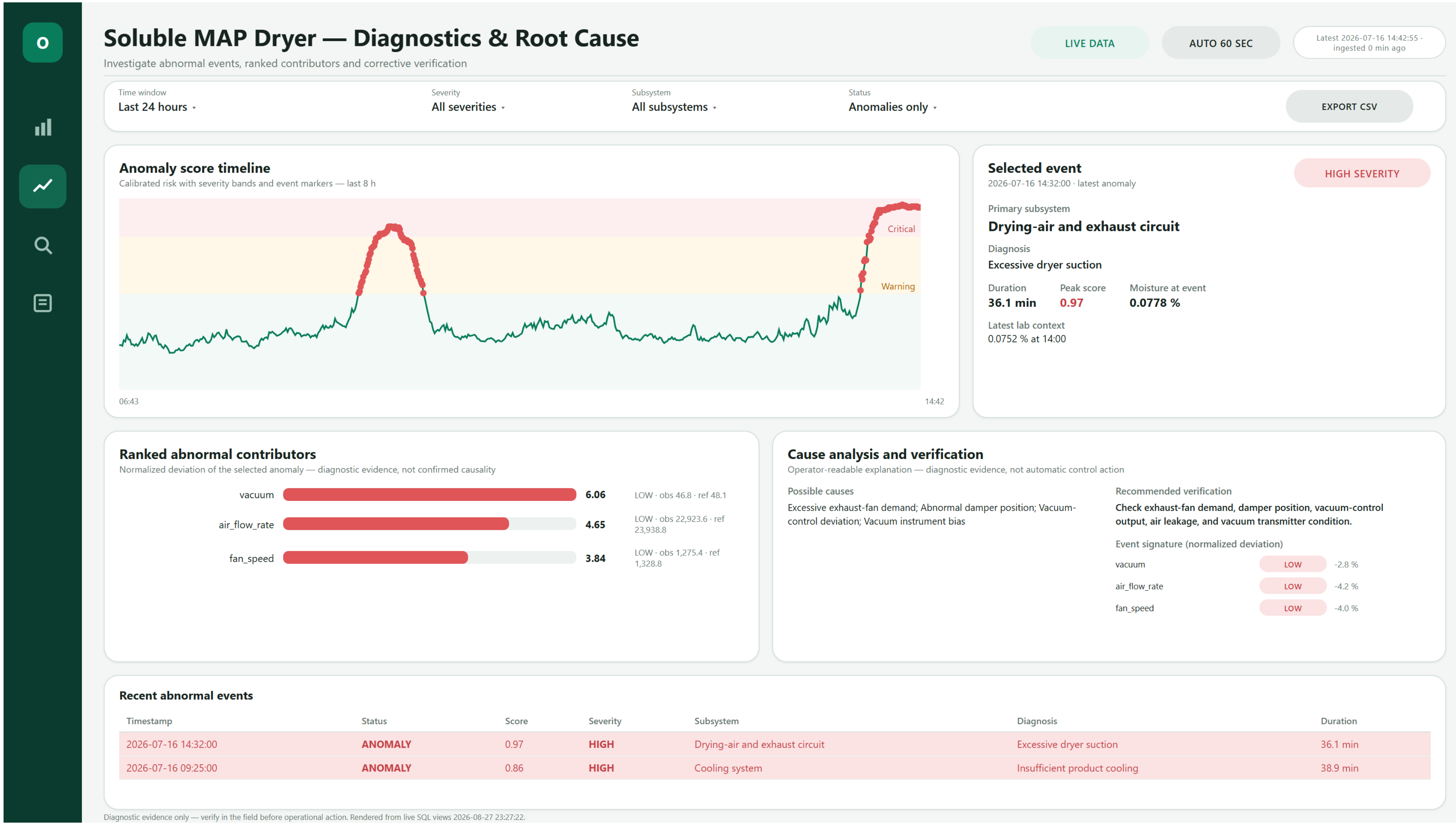


Figure 7.2: Final Diagnostics capture with event selection, ranked contributors and verification guidance.
Source: Current SQL-backed full-screen dashboard capture; Author's work.

The captured high-severity demo event points to the drying-air and exhaust circuit. Vacuum, air flow and fan speed are all low relative to the reference profile, so the displayed checks focus on fan demand, damper position, air leakage and vacuum measurement. This is a probable diagnosis for the generated scenario, not proof that one component failed.

Consecutive anomalous rows separated by no more than 30 seconds are grouped into one candidate event. Grouping reduces row-level repetition, but it does not turn a statistical novelty into an approved process alarm. Alarm management requires governed priorities, ownership, response procedures, performance monitoring and alignment with the site philosophy [17]. The present event list is an advisory investigation aid only.

7.4 Why the demonstration source is separate

A short jury session cannot rely on the chance that a useful held-out event happens at the right moment. I therefore prepared a two-day demonstration fork with eight disclosed injected episodes. The source preserves normal rows and sparse laboratory values, and the frozen models are not retrained on the injections.

Across that fork, the model flags 0.071% of non-injected rows and 58.7% of injected rows; seven of the eight injected episodes receive a model response. These numbers describe presentation coverage, not independent novelty-model validation. The canonical TEST analysis in Chapter 5 remains the model evidence and includes the feed-surge miss.

Table 7.1: Dashboard-demo audit and correct interpretation

Item	Result	Meaning
Source	34,560 rows / 2 days	Dedicated presentation replay.
Injected scenarios	8 episodes	Disclosed, deterministic demonstrations.
Normal-period model flags	0.0706%	Behaviour on non-injected demo rows; not a plant false-positive rate.
Injected-period model flags	58.73%	Visibility during scripted intervals; not plant detection performance.
Episode response	7 of 8	Seven demo episodes produce at least one model flag.

Source: resources/dashboard_demo/MAP_Dryer_Dashboard_Demo_Audit.json.

7.5 Usability and presentation performance

The final launcher opens the report in a PowerPoint-like full-screen presentation workflow so the jury sees the canvas rather than the authoring panes. Page navigation remains manual, but the data state refreshes automatically. Shortening the refresh request to two seconds can make switching pages less smooth because DirectQuery requests and visual rendering overlap. For the defence, stable five-second or conservative presentation refresh is more credible than an aggressive interval that produces lag.

Industrial use would require operator testing, accessible colour and text cues, approved terminology, nuisance-event review and a clear stale-data procedure. It would also require gateway and capacity measurements under expected concurrency. The current pages demonstrate the information architecture and the software connection; they do not establish a validated human–machine interface.

CHAPTER 8

Results, Validation, Limitations and Future Work

8.1 Consolidated result

The project now forms one inspectable chain: supplied CSV, causal preprocessing, frozen model artifacts, Python inference, versioned PostgreSQL records and a two-page DirectQuery dashboard. The strongest evidence is not simply that each component runs. It is that the model advantages and failures, source boundaries and interface responsibilities can all be traced to a specific artifact.

Table 8.1: Consolidated quantitative evidence

Evidence	Result	Correct interpretation
Ridge TEST	MAE 0.001069; RMSE 0.001403; R^2 0.824530; $n = 165$	Untouched synthetic chronological tail; percentage-point errors.
Persistence baseline	MAE 0.001459; RMSE 0.001848	Ridge improves MAE 26.8% and RMSE 24.1%.
Residual audit	Bias 0.000066; Durbin–Watson 1.827; lag-1 autocorrelation 0.084	Limited short-lag structure on this TEST sequence; not a plant stability claim.
One-Class SVM TEST	2.654% total flags; 60.25% labelled-row response; 2.258% normal-period flags	Process-novelty response on synthetic labelled episodes.
Episode response	Steam 82.0%; airflow 79.3%; feed surge 0%	Sensitivity varies by departure type; the loading episode needs further work.
Full analytical replay	237,600 rows in 2.037 s; about 116,626 rows/s	Batch model/feature computation only; database and Power BI excluded.
Database smoke test	1,500 rows; 9/47 ms average/maximum cycle latency	Bounded integration evidence, not production load qualification.
Dashboard demo	34,560 rows; 8 injections; 7/8 episode response	Jury presentation coverage only.

Source: Frozen artifacts and reproducible report audit; detailed evidence in Chapters 5–7.

8.2 Verification performed on the final repository

The current Python suite passes 77 tests plus 8 subtests. It covers timestamp and cadence contracts, causal alignment, feature order, artifact inference, replay behaviour, diagnosis rules, persistence, database helpers and presentation utilities. The Power BI report validator resolves 114 field references across 86 visuals against 303 model fields in six tables, with a 1600×900 canvas.

A live PostgreSQL contract check also passes for all six DirectQuery views. At the time of the final check, the database contained a short 41-row current replay fragment, one laboratory row and no anomaly/contributor event rows. Empty optional event views are valid in a normal interval. This live check proves column compatibility and connectivity at that moment; it is not a substitute for the larger analytical or smoke-test results.

8.3 What I consider demonstrated

The work demonstrates that sparse laboratory values can be preserved while a faster process stream supports advisory inference. It shows that residence-aware alignment and strictly previous laboratory context can be implemented without obvious future leakage. It also shows a measurable soft-sensor improvement over a realistic persistence baseline.

Table 8.2: Validation layers and their remaining boundary

Layer	What passed	What it does not prove
Unit and contract tests	77 tests + 8 subtests	Plant representativeness or operator acceptance.
Report semantic validation	86 visuals and all 114 field references resolve	Query latency under deployment concurrency.
Live SQL / Tabular Model Definition Language (TMDL) check	All six views match the semantic model	Long-duration resilience or gateway behaviour.
Model holdout	Chronological soft-sensor and synthetic-scenario anomaly audit	Accuracy on new OCP grades, seasons or real failures.
Presentation replay	Full-screen two-page flow with deterministic events	Independent model validation.

The anomaly result is deliberately narrower. The frozen model responds strongly to two held-out disturbance families and does not flag the feed surge. What is demonstrated is an auditable novelty-to-guidance workflow, not comprehensive fault classification.

Finally, the complete software chain is coherent. The exact notebook artifacts reach the runtime, event records are idempotent and versioned, SQL views match the semantic model, and the dashboard preserves measurement/prediction semantics. This is an Industrial Engineering contribution because data quality, timing, interpretation and operator authority are treated as one system.

8.4 Material limitations

- **Synthetic source.** The supplied file supports reproducible prototype testing, but it cannot establish performance across real grades, seasons or operating modes.
- **No live plant interface.** PCS 7 screens informed the analysis; CSV replay substitutes for historian and laboratory acquisition.
- **Small supervised evidence.** The dense stream has 1,589,760 rows, but only 1,103 usable moisture targets. Label cadence, not process-row count, governs the supervised evidence.
- **Scenario-dependent novelty detection.** The detector reacts to steam and airflow departures but needs representative plant history and confirmed events for calibration.
- **Demonstration injections.** The two-day dashboard fork improves presentation coverage but cannot add credibility to model validation; it is disclosed and kept separate.
- **Incomplete physical balance.** Wet-cake moisture, dry-solids flow, gas humidity/enthalpy and heat loss are unavailable, so thermal features remain proxies.
- **Plant tag confirmation.** The canonical vacuum field follows the supplied convention; its polarity and pressure reference require confirmation.
- **Prototype timing.** Five-second event/replay cadence and local performance do not qualify historian, network, gateway, concurrent Power BI or failover performance.
- **No alarm or control authority.** Statistical thresholds are not approved operating, alarm, trip or safety limits, and no actuator command exists.

8.5 Stage-gated industrialization roadmap

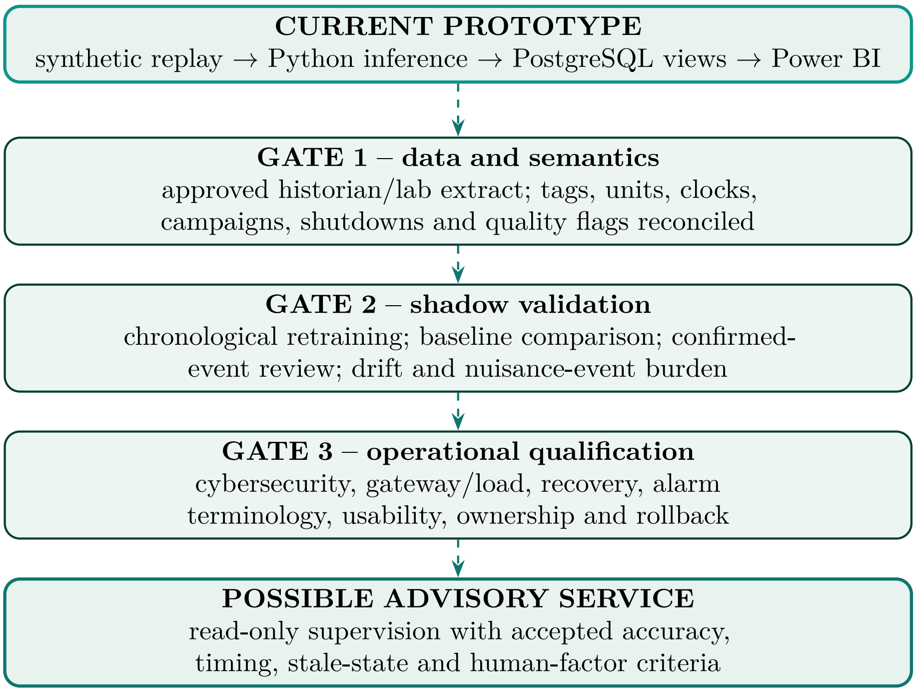


Figure 8.1: Evidence gates from the current prototype to a possible industrial advisory service.
Source: Author's work.

The next step should be data, not control. An approved read-only extract should cover normal campaigns, grade changes, shutdowns, maintenance and instrument-quality states, with laboratory sample and result times distinguished. The soft sensor should be retrained and compared by campaign against persistence and simple process baselines. The prototype currently provides point predictions; formal uncertainty estimation could be evaluated later using representative plant data.

For anomaly detection, confirmed events and normal operating modes are more valuable than another isolated hyperparameter search. A shadow review can label whether an event was actionable, expected, caused by data quality or unresolved. Those labels would support threshold selection, alternative semi-supervised models and a credible nuisance-event measure.

Only after data, model and service qualification should OCP consider a persistent advisory deployment. Any later recommendation layer would still require explicit operator authority and approved procedures. Autonomous control is outside the present roadmap.

8.6 Personal contribution summary

My main contribution was to connect disciplines that are often presented separately. I converted process observations into a defensible data contract, implemented causal alignment, compared the soft sensor against a practical baseline, exposed the anomaly model's missed scenario, built the runtime and database boundary, and shaped the dashboard around operator verification. I also added the evidence audit and report traceability needed to defend both the result and its limits. The remaining work is not hidden technical polish; it is the industrial evidence required before the prototype can leave a controlled environment.

General Conclusion

This internship began with a two-hour information gap and ended with a working, traceable supervision chain. I translated the dryer problem into a multi-rate data contract, built causal process-to-laboratory alignment, compared moisture models chronologically, connected the selected artifacts to PostgreSQL and designed a two-page Power BI interface. The result does not replace laboratory control or PCS 7. It gives the operator an additional, explicitly advisory view between confirmed samples.

The soft-sensor branch is the strongest quantitative result. On the untouched 165-sample TEST tail, Ridge achieves MAE 0.001069 and RMSE 0.001403 percentage points H_2O, improving on a last-laboratory persistence baseline by 26.8% and 24.1%, respectively. Its R^2 is 0.824530 and its mean bias is small. The model remains advisory and its prediction stays distinct from the laboratory measurement.

The anomaly branch provides earlier process awareness rather than fault classification. On the held-out canonical TEST interval, it flags 60.2% of labelled synthetic disturbance rows, responding strongly to steam-dip and airflow-restriction episodes but not to the feed surge. The implemented value lies in the complete path from novelty score to ranked contributors, likely subsystem, possible causes and recommended verification. Representative plant history would be needed to recalibrate the boundary and extend its sensitivity.

The dashboard demonstration uses a separate two-day fork with eight disclosed injected episodes so that the jury can observe state changes during a short session. That fork is presentation evidence, not a model benchmark. Power BI only queries and displays PostgreSQL views; Python performs preprocessing, inference and diagnostic transformation. Laboratory moisture remains the quality reference throughout.

The final classification is therefore precise: the project is a reproducible engineering prototype, not a plant-ready system. Its next credible step is a read-only shadow study using approved historian and laboratory data, followed by campaign-based validation, operator review, alarm governance, cybersecurity qualification and explicit acceptance criteria. The contribution is not an isolated algorithm. It is a transparent path from process data to a decision-support interface, including clear evidence of what currently works and what still needs to be proven.

CHAPTER A

Evidence and Reproduction Appendices

A.1 Claim-to-artifact traceability

Table A.1: Principal claims and reproducible evidence

Claim	Primary artifact	Independent report check
Canonical row count, cadence and sparse lab count	`data/processed/MAP_Dryer_Canonical_5s.csv`	`report_enhancement_metrics.json` and direct timestamp audit.
Causal 16-input moisture contract	`MAP_Dryer_Lab_Aligned_16.csv`; `feature_schema.json`	First lab row excluded; strictly previous laboratory lookup verified by tests.
Ridge selection and TEST metrics	`notebook03_model_evaluation.json`; frozen pipeline	Recomputed predictions, persistence baseline and residual audit.
One-Class SVM configuration	`notebook04_anomaly_evaluation.json`; frozen model/scaler	Validation flag rate, held-out episode response and artifact provenance.
Dashboard-demo disclosure	`MAP_Dryer_Dashboard_Demo_Audit.json`	Normal/injected rows, lab preservation and episode response checked.
Runtime/database contract	`realtime_service.py`; SQL bootstrap/views	Bounded smoke log and live six-view TMDL comparison.
Power BI field integrity	Power BI report and semantic-model definitions	86 visuals, 114 references, 303 fields, six tables.

Key SHA-256 identifiers at the time of this report are: canonical CSV `AE9216518E73...`, moisture artifact `B525C7B8598E...`, anomaly model `07BCE32B6BF7...` and anomaly scaler `AF28219037FD.` ... Full hashes remain in the manifest, model metadata and repository audit output.

A.2 Source-label audit

Table A.2: Manifest catalog versus direct canonical label audit

Evidence	Count	Treatment in this report
Manifest disturbance catalog	7 entries	Retained as supplied metadata; identified as stale/incomplete.
Distinct embedded scenario episodes	22 episodes	Used for actual counts after direct timestamp/label audit.
Episodes in held-out TEST	3 episodes	Used for frozen-detector scenario response.

A.3 Reproduction commands

The report-specific audit is read-only with respect to canonical data and model artifacts. From the project root:

```
$env:PYTHONPATH="src"
python tools/generate_report_enhancement_evidence.py
python -m pytest -q
python tools/validate_report_fields.py
python realtime_pipeline/src/verify_powerbi_views.py
```

The first command regenerates the JavaScript Object Notation (JSON) metrics audit, the soft-sensor evidence, the anomaly holdout evidence and the synchronized dashboard captures. The last command requires the configured local PostgreSQL instance. The report itself is compiled repeatedly with `pdflatex` so references and contents are resolved.

A.4 Principal paths

Table A.3: Minimum artifact set needed to inspect the reported workflow

Path	Purpose
`final_report/evidence/report_enhancement_metrics.json`	Independent numerical audit used in the enhanced report.
`artifacts/notebook03_model_evaluation.json`	Chronological soft-sensor selection and metrics.
`artifacts/notebook04_anomaly_evaluation.json`	One-Class SVM export provenance and validation flag rate.
`models/5s/feature_schema.json`	Ordered moisture and anomaly feature contracts.
`resources/dashboard_demo/*Audit.json`	Disclosure and response audit for jury-only injections.
`realtime_pipeline/src/realtime_service.py`	Replay, inference, diagnosis and transactional persistence.
`POWERBIDASHBOARD/*.pbip`	Power BI project entry point.

Bibliography

[1] OCP Group, “Who We Are,” official website, accessed 8 Aug. 2026. [Online]. Available: `https://www.ocpgroup.ma/en/who-we-are`

[2] OCP Group, “Industrial Operations,” official website, accessed 8 Aug. 2026. [Online]. Available: `https://www.ocpgroup.ma/en/what-we-do/industrial-operations`

[3] Anonymous student author, *Internship Report* [title translated from French], supplied ENSAM/OCP report in Microsoft Word Open XML Document (DOCX) format, contextual reference consulted during project preparation, undated.

[4] K. Souilmi, *Creation of a ProcessBook for Soluble MAP Training* [title translated from French], internship report, OCP Group / ENSAM, academic year 2018–2019, supplied as a Portable Document Format (PDF) file, 40 pp.

[5] C. Abdelkarim, “Internship field observations on soluble MAP production and PCS 7 supervision,” personal notes, OCP Jorf Lasfar, July–August 2026. Notes consulted as source material; images are not reproduced.

[6] C. Abdelkarim, *MAP-Dryer-AI repository: five-second multi-rate supervision prototype*, source code and README, local project revision audited 28 Aug. 2026.

[7] MAP-Dryer-AI, “Data dictionary and subsystem feature metadata,” `config/data_dictionary.yaml` and `config/feature_metadata.yaml`, audited 8 Aug. 2026.

[8] MAP-Dryer-AI, “Notebook 03: Model 1—Soft Sensor for Final Moisture,” `notebooks/03_Model1_SoftSensor.ipynb` and `artifacts/notebook03_model_evaluation.json`, executed and audited 24 Aug. 2026: Ridge, $\alpha = 10$, chronological TEST $R^2 = 0.824530$ ($n = 165$).

[9] MAP-Dryer-AI, “Notebook 04: Model 2—Anomaly Detection and Diagnosis,” `notebooks/04_Model2_AnomalyDetection&Diagnosis.ipynb` and `artifacts/notebook04_anomaly_evaluation.json`, executed and audited 24 Aug. 2026: 15-feature One-Class SVM runtime export with 5.925% validation flag rate.

[10] MAP-Dryer-AI, “PostgreSQL base schema and five-second upgrade,” `POWERBIDASHBOARD/sql/bootstrap_base_schema.sql` and `upgrade_5s_schema.sql`, audited 8 Aug. 2026.

[11] MAP-Dryer-AI, “Bounded held-out TEST replay log,” `artifacts/realtime_90s_stdout.log`, 1,500 source rows, 24 Aug. 2026.

[12] P. Kadlec, B. Gabrys and S. Strandt, “Data-driven soft sensors in the process industry,” *Computers & Chemical Engineering*, vol. 33, no. 4, pp. 795–814, 2009. doi: 10.1016/j.compchemeng.2008.12.012.

[13] B. Schölkopf, J. C. Platt, J. Shawe-Taylor, A. J. Smola and R. C. Williamson, “Estimating the support of a high-dimensional distribution,” *Neural Computation*, vol. 13, no. 7, pp. 1443–1471, 2001. doi: 10.1162/089976601750264965.

[14] C. Bergmeir and J. M. Benítez, “On the use of cross-validation for time series predictor evaluation,” *Information Sciences*, vol. 191, pp. 192–213, 2012. doi: 10.1016/j.ins.2011.12.028.

[15] V. Venkatasubramanian, R. Rengaswamy, K. Yin and S. N. Kavuri, “A review of process fault detection and diagnosis: Part I,” *Computers & Chemical Engineering*, vol. 27, no. 3, pp. 293–311, 2003. doi: 10.1016/S0098-1354(02)00160-6.

[16] M. Rousselet and V. K. Dhir, “Numerical modeling of a co-current cascading rotary dryer,” *Food and Bioproducts Processing*, vol. 99, pp. 166–178, 2016. doi: 10.1016/j.fbp.2016.05.001.

[17] International Electrotechnical Commission, *IEC 62682:2022 – Management of alarm systems for the process industries*, 2nd ed., 2022. [Online]. Available: `https://webstore.iec.ch/en/publication/65543`

[18] Microsoft, “Automatic page refresh in Power BI,” Microsoft Learn, accessed 8 Aug. 2026. [Online]. Available: `https://learn.microsoft.com/en-us/power-bi/create-reports/desktop-automatic-page-refresh`